

# *Spooky Computing*

How Quantum Mysteries are Shaking  
the Foundations of Computer Science

NKU CENSAM TALK - November 13, 2001

Kevin G. Kirby

Department of Mathematics and Computer Science

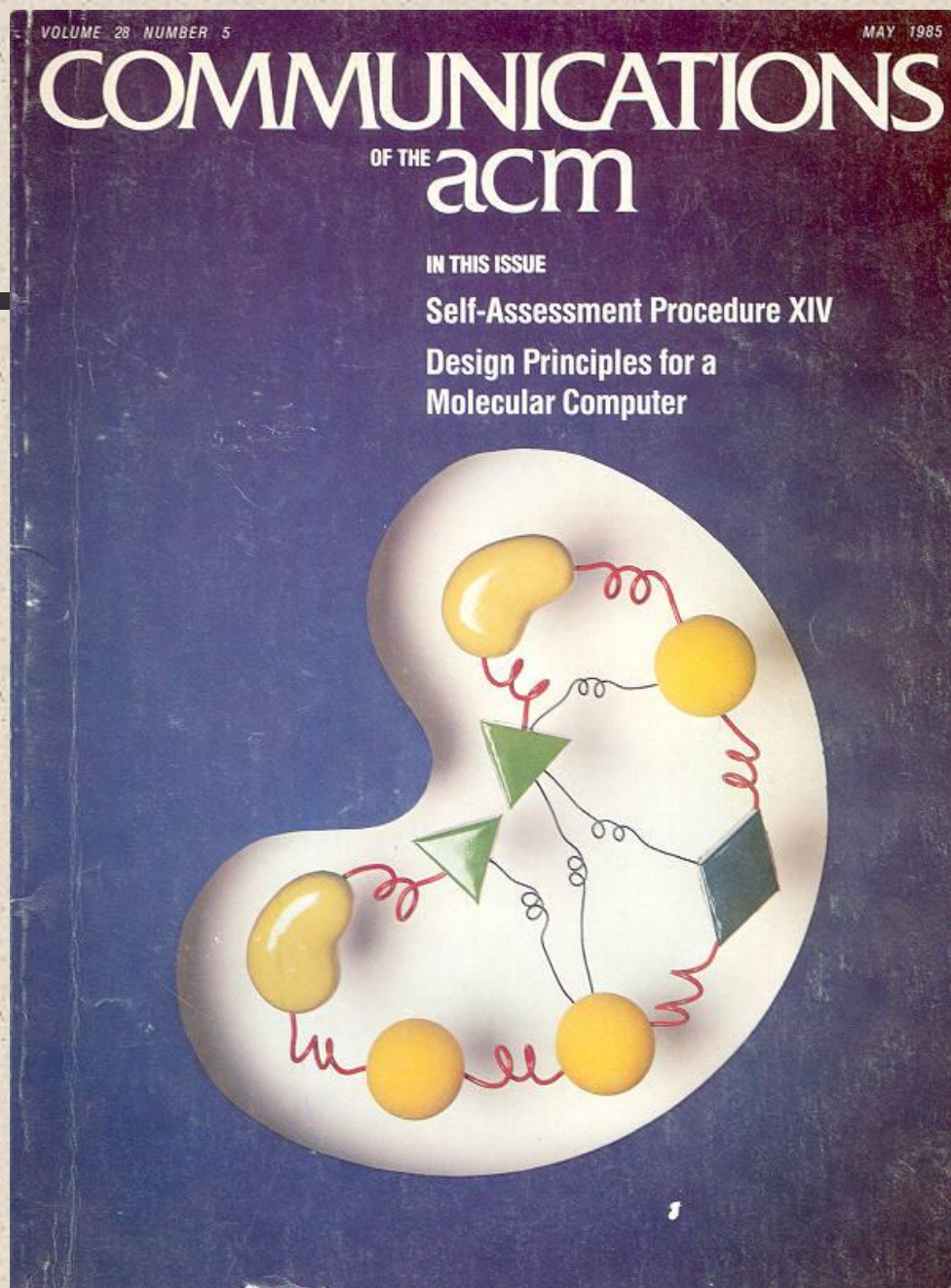


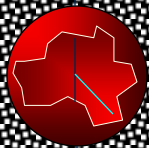


## Michael Conrad 1940 - 2000

Earliest Artificial Life Programs (1969)  
Early “Neural Darwinism” model (1974)  
Adaptability theory (1972-1983)  
Cytoskeletal computing models (1986)  
Theories of biocomputing (1979-2000)  
Quantum mechanics / life (1993-2000)

Computer science as  
*natural science.*

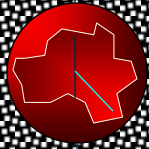




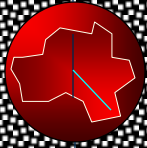
# Outline

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1. Background
2. The cartoon overview
3. One qubit
4. Several qubits
5. Some quantum gates, circuits and algorithms
6. Quantum database search
7. Appendix: Spinor physics
8. Appendix: QC *vs.* BMC and NNC



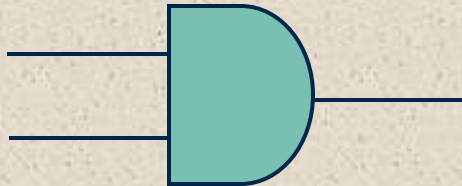
# 1 Background



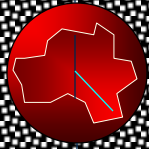
# Prerequisites?

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You don't need to know Ohm's law to understand a logic gate.



Just a little boolean algebra.



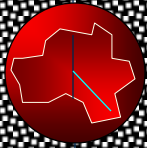
# Prerequisites?

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You don't need to know Schrödinger's equation to understand an quantum gate.



Just a little **linear** algebra.



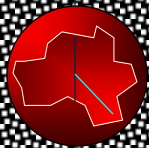
Why bother?

## Quantum Computing

Since current semiconductor technology is reaching its limits, we need a new technology so we can make **faster and smaller** computer components.

*This is what quantum computing is all about.*

***No!***



Why?

# Quantum Computing

Since current semiconductor

tech

lin

techn

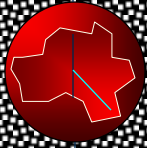
faster

T

co

***We get...***

- ***A new understanding of the nature of information & computation***
- ***Fundamentally faster algorithms***  
[faster when implemented on quantum hardware]



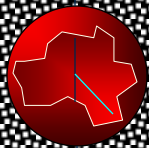
## Quantum Computing .NET ?

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[MS Research founder Rick] Rashid refers to the whole of Microsoft's research efforts as the "Star Trek" division of Microsoft because they are working on somewhat futuristic technologies such as nanotechnology, virtual collaboration, and language translation.

Microsoft Research, Rashid said, has even published a paper on **quantum state teleportation**.

-- "Peering into future software research"  
*InfoWorld*, September 5, 2001



## Quantum Computing NET 2

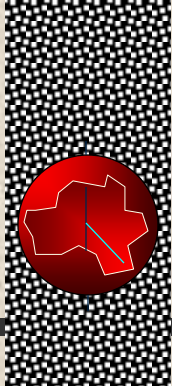
**“I feel strongly that papers on quantum teleportation are beneath contempt.”**

[MS Research efforts as the somewhat full and language translation.

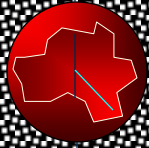
--- W.E. Lamb Jr. [Nobel Laureate]  
*Am. J. Phys.* 69 (4), April 2001

Microsoft Research, Rashid said, has even published a paper on **quantum state teleportation**.

-- “Peering into future software research”  
*InfoWorld*, September 5, 2001

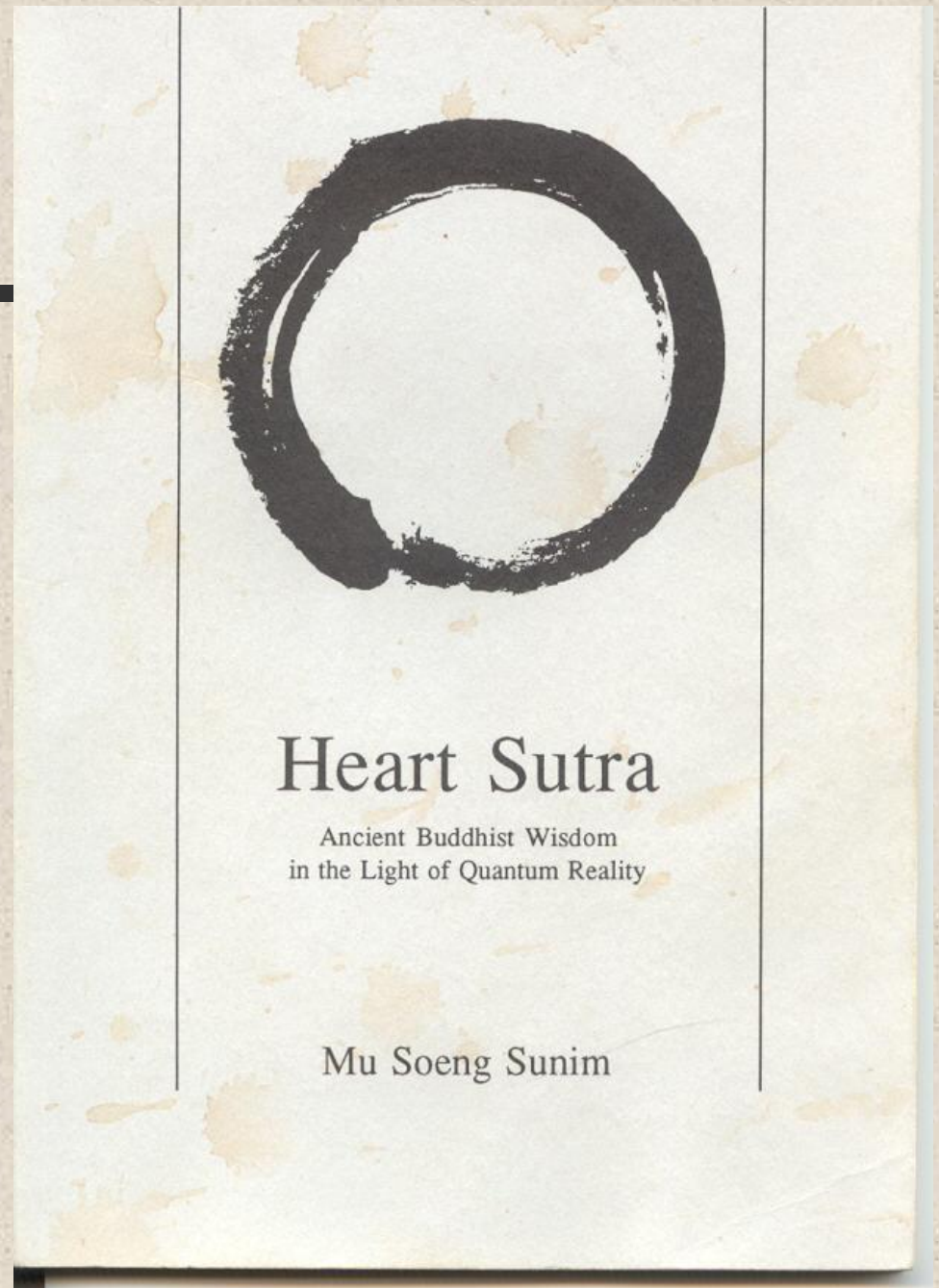


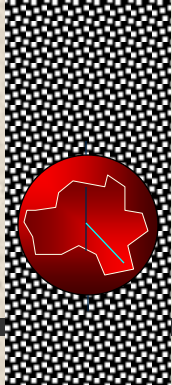
The origins: Wolfgang Pauli, May 1927: Spin (2-level systems)



# Quantum mysteries

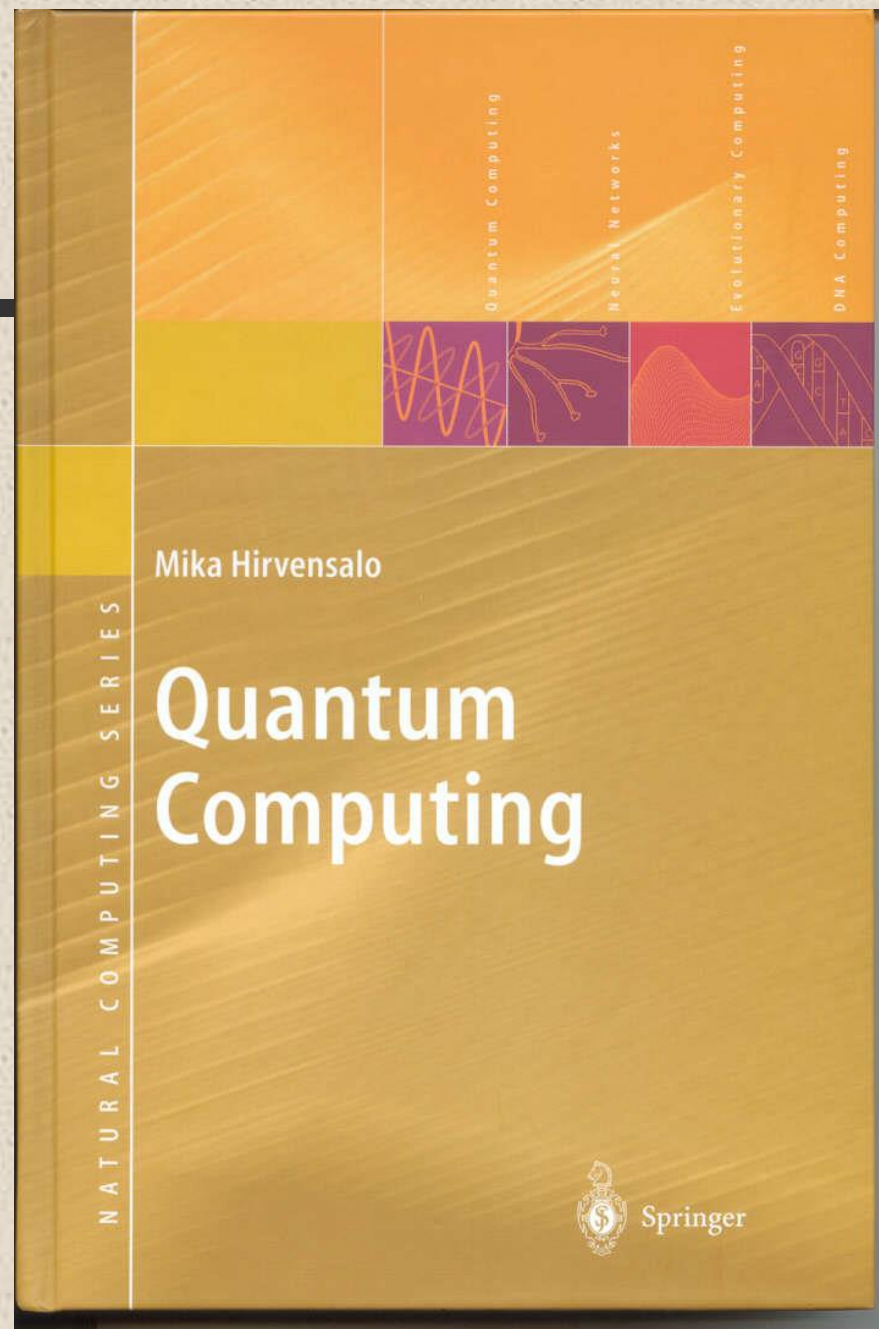
1. Superposition
2. “Spooky action at a distance”

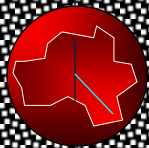




# Quantum mysteries

1. Superposition
2. “Spooky action at a distance”





# Some Beginnings

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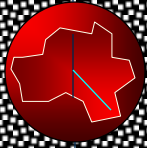
1925 Quantum mechanics

1980 Quantum computability

1994 Quantum fast factoring

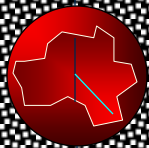
1996 Quantum fast search

2000 A simple 7-qubit quantum computer



# 2

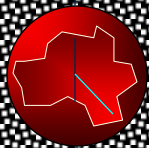
## The Cartoon Preview



# The Bit

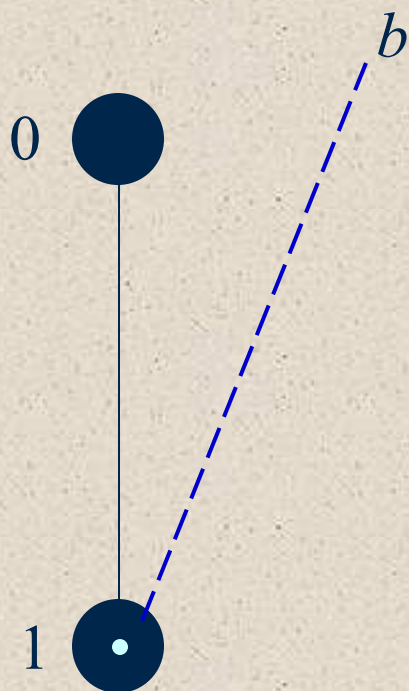
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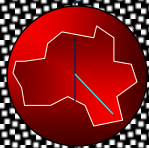


# The Bit

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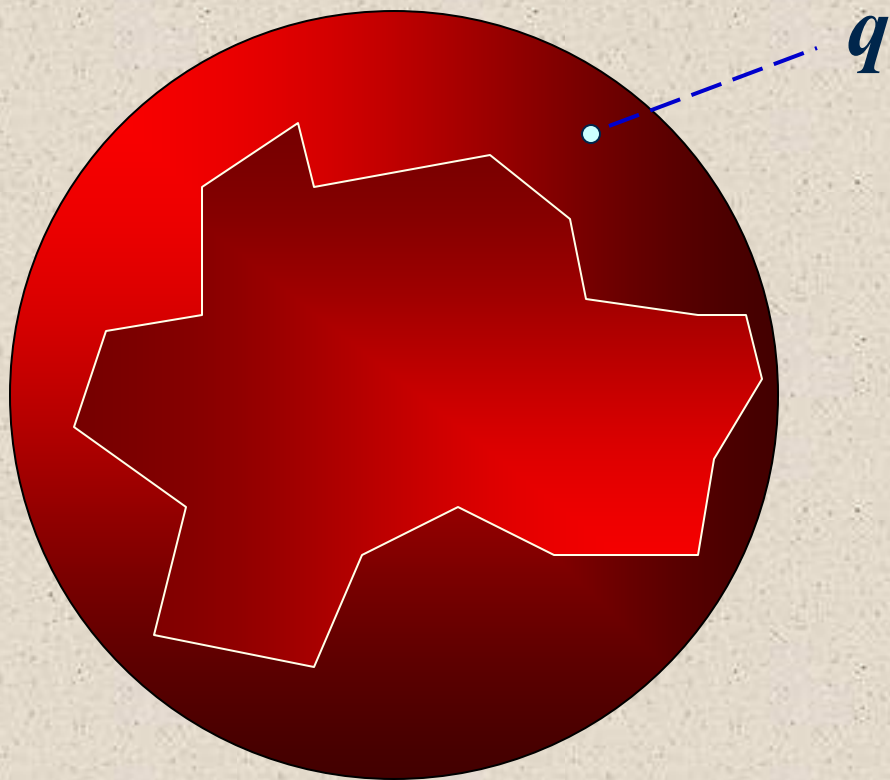


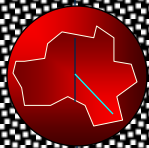
2 possible values



# The Qubit

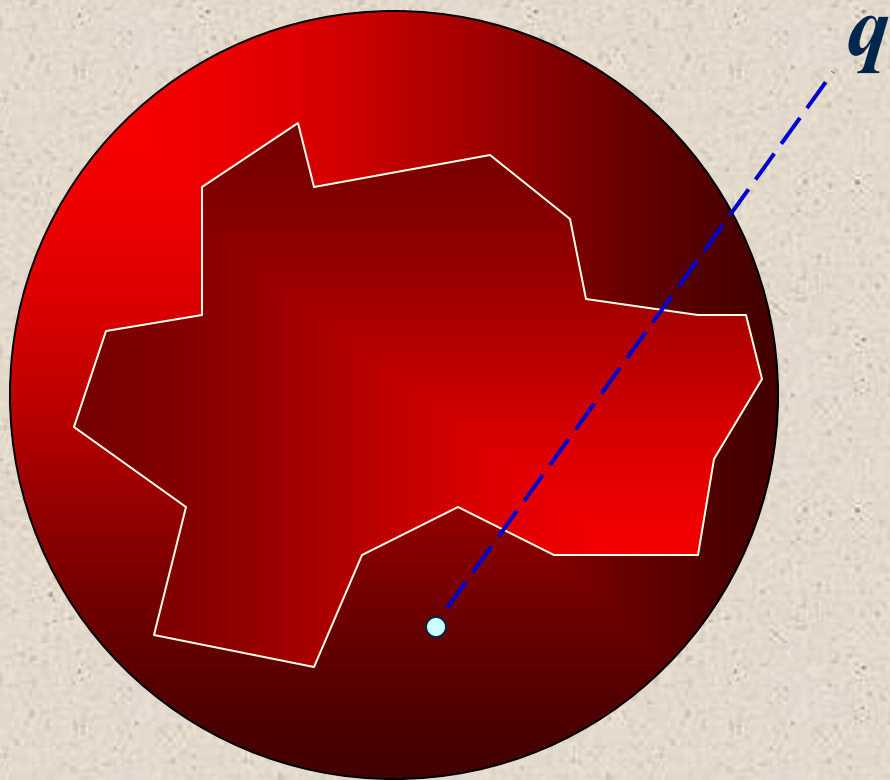
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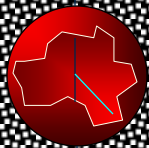


# The Qubit

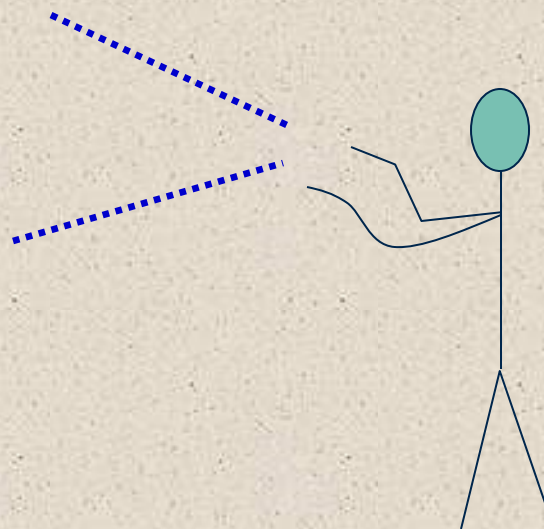
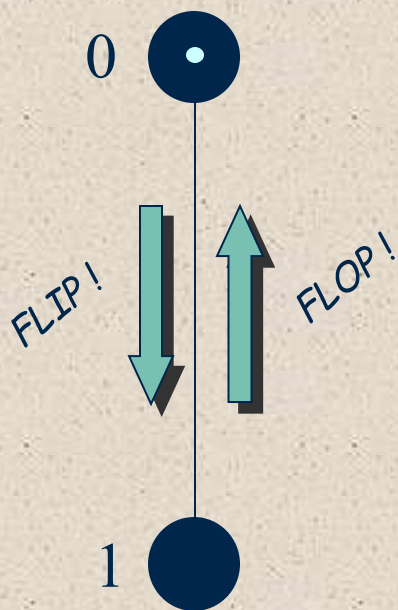
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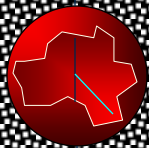
$2^{\aleph_0}$  ( $\infty$ ) possible values



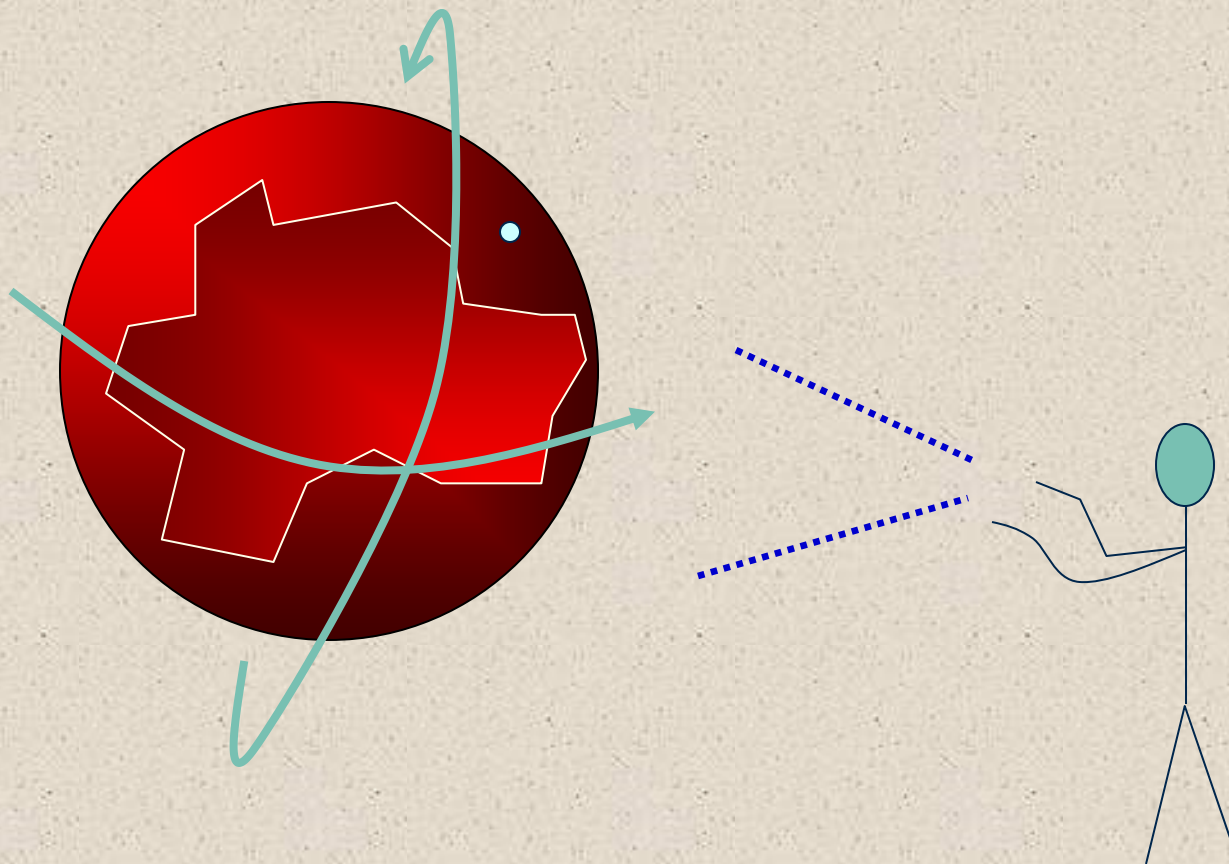
# Changing a Bit



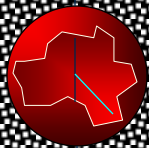
*Negate*



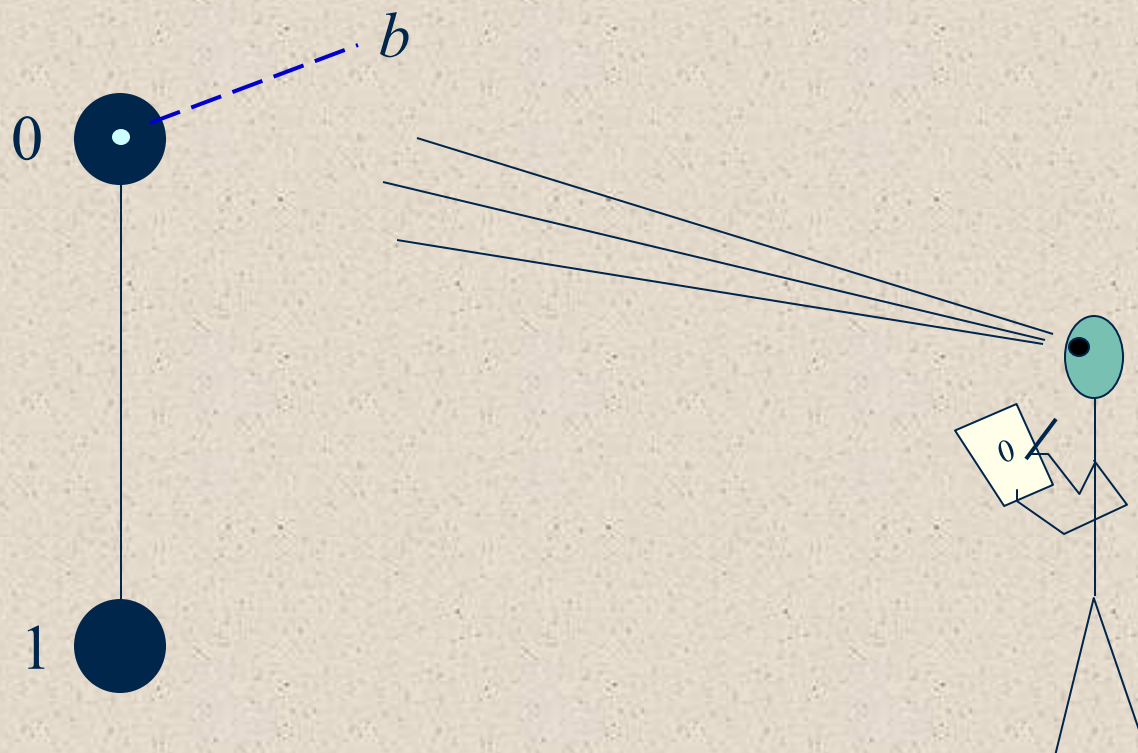
# Changing a qubit



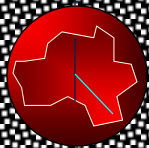
*Rotate, reflect, ...*



# Reading a Bit

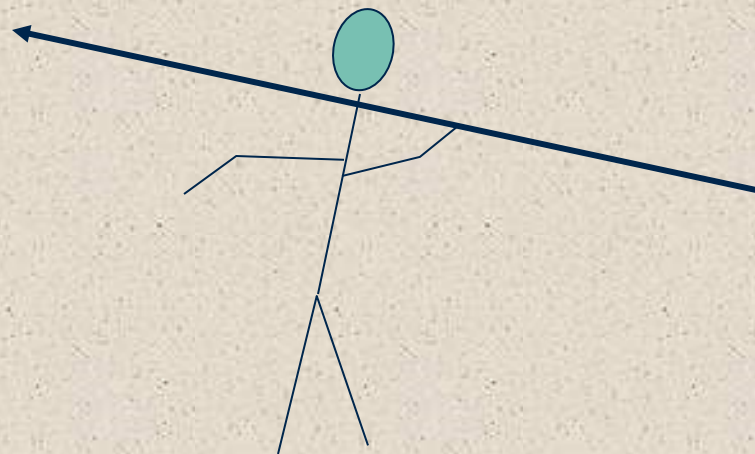
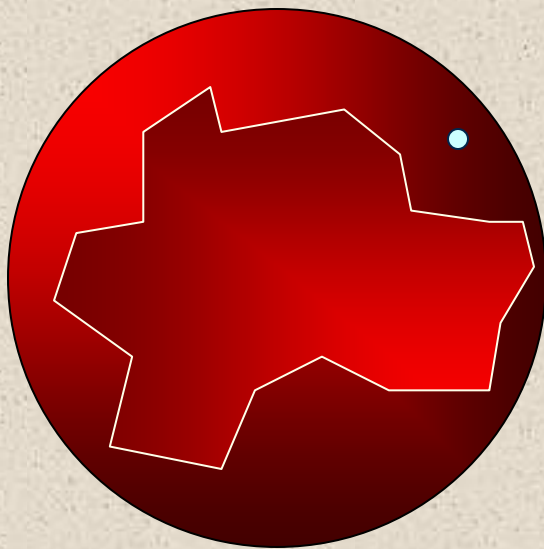


1. *Perfect fidelity.*
2. *No side effects.*

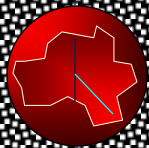


# Reading a qubit [1]

---

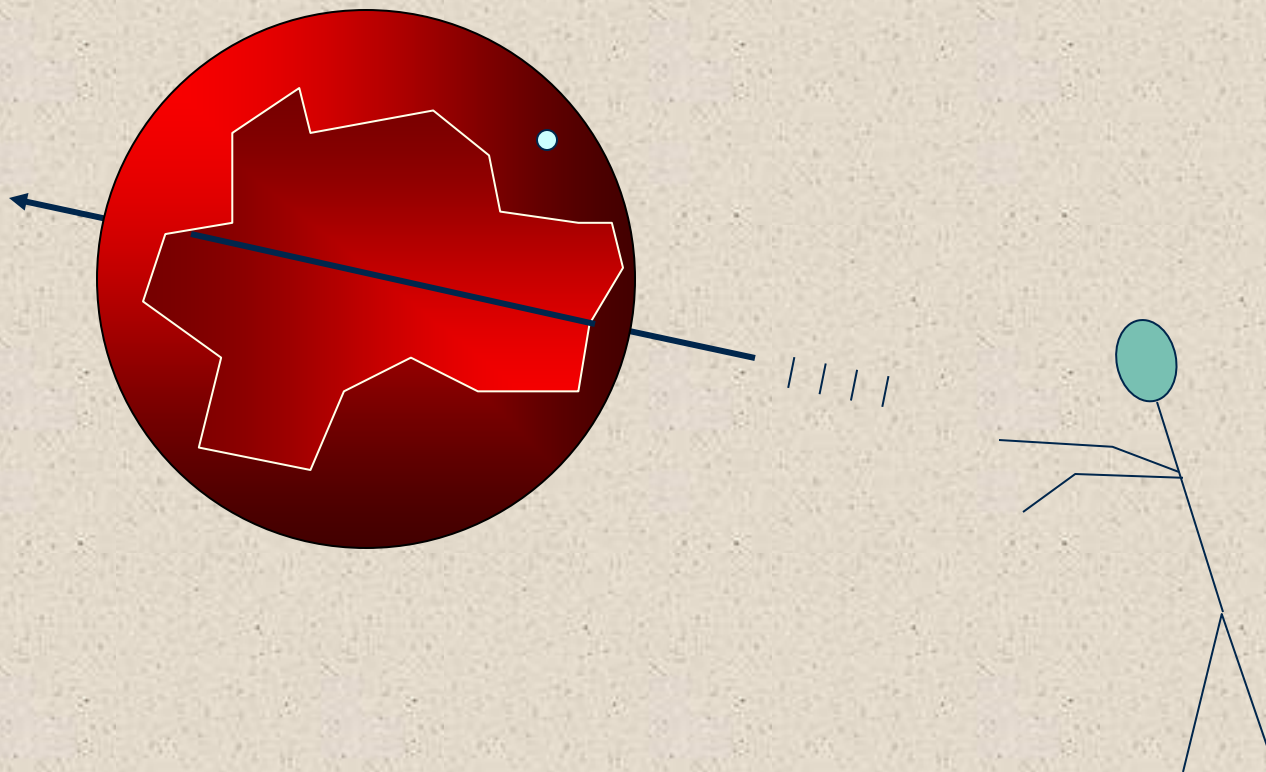


*Choose a “readout basis”.*

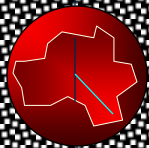


# Reading a qubit [2]

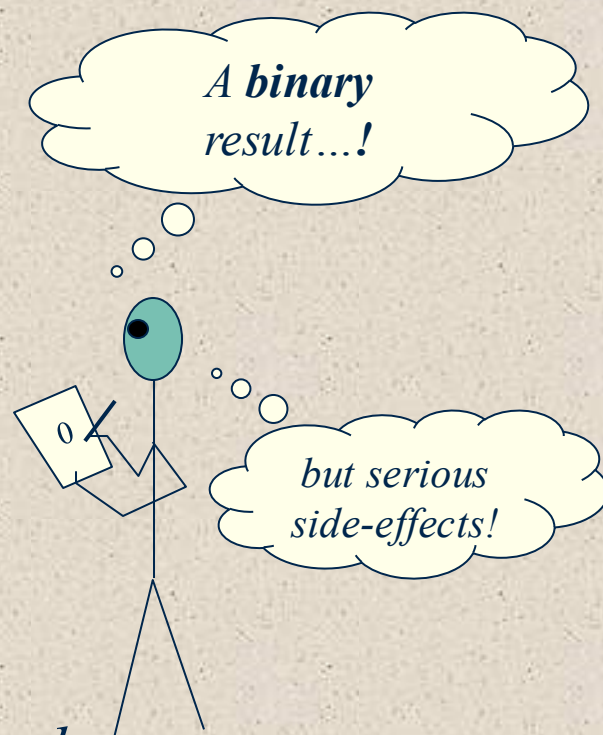
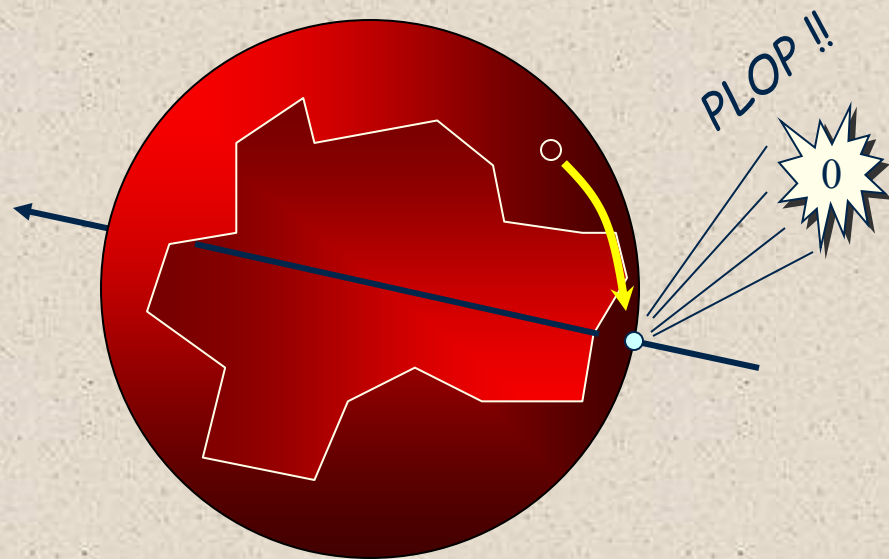
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*Measure!*



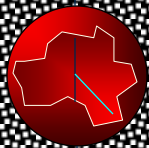
# Reading a qubit [3.0]



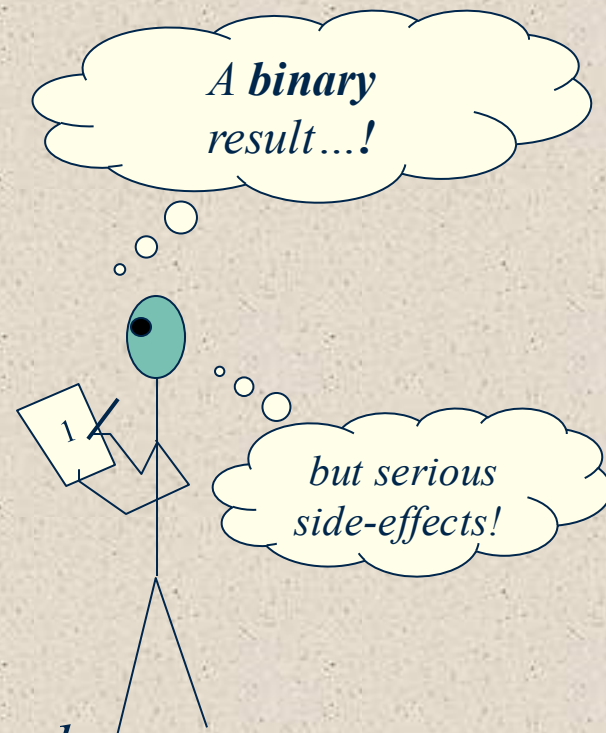
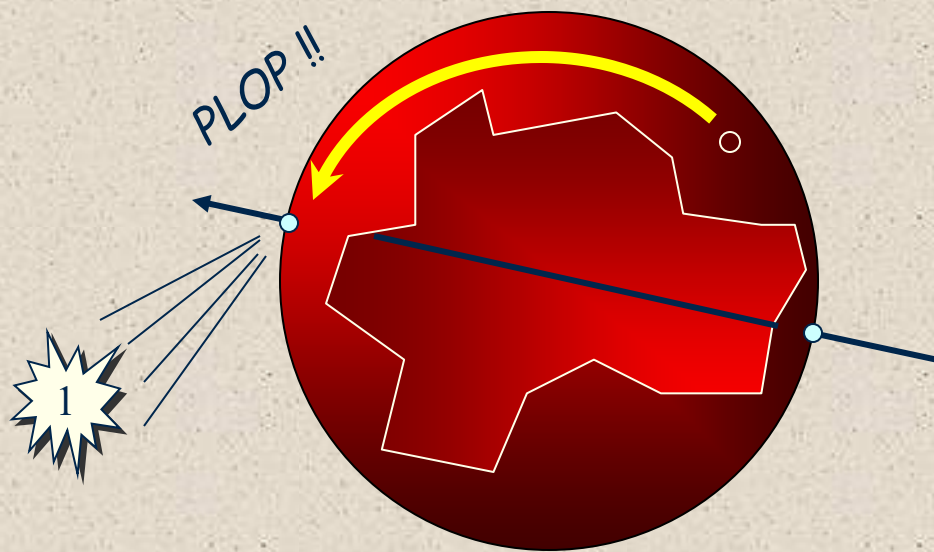
*Qubit makes a random jump to one of the two ends.*

*(The closer it was to an end, the more likely it was to jump there.)*

*You can't find out what the original spot really was.*



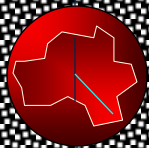
# Reading a qubit [3.1]



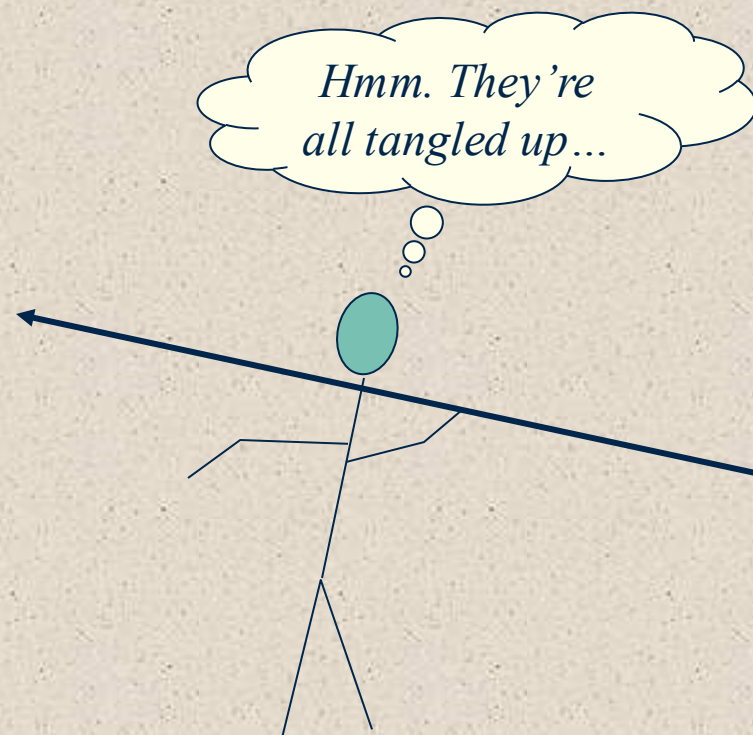
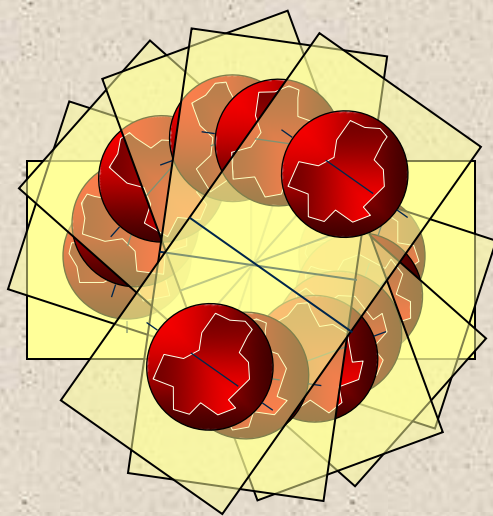
*Qubit makes a random jump to one of the two ends.*

*(The closer it was to an end, the more likely it was to jump there.)*

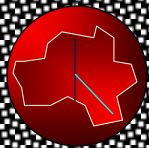
*You can't find out what the original spot really was.*



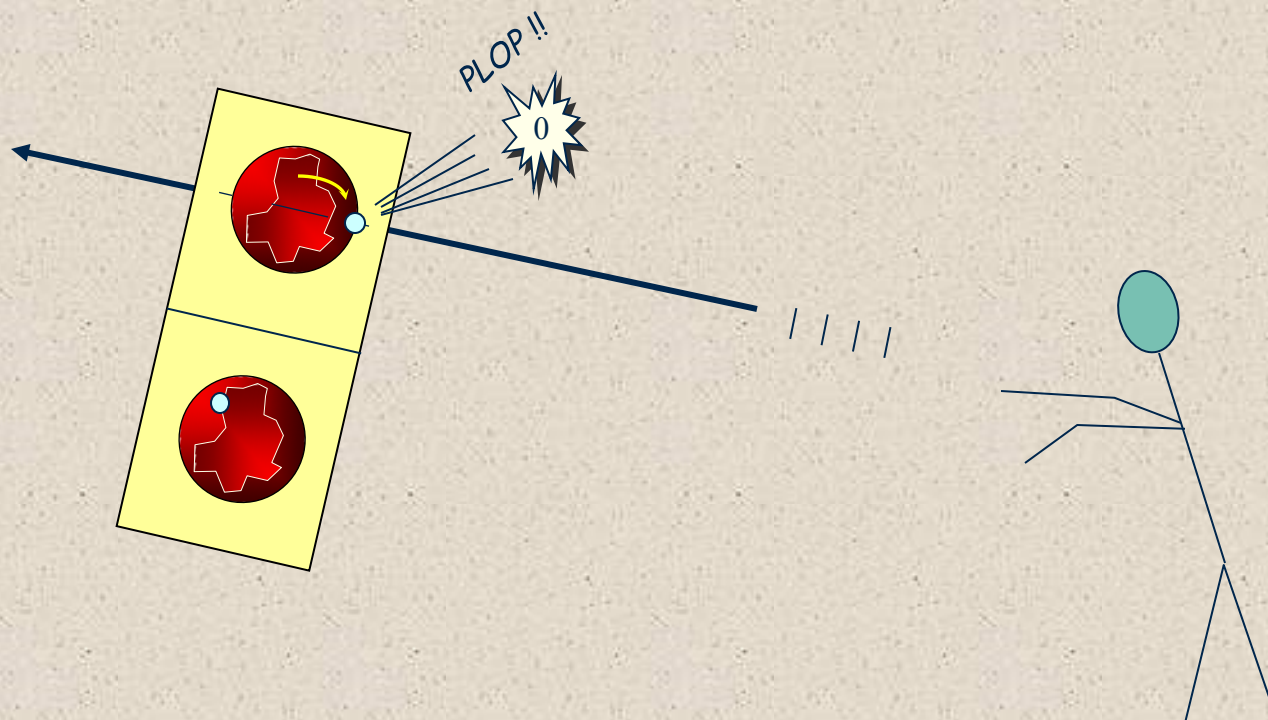
# Reading a 2-qubit register [1]



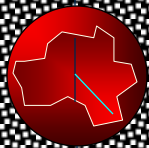
*Pick which qubit you want to read; choose your readout basis...*



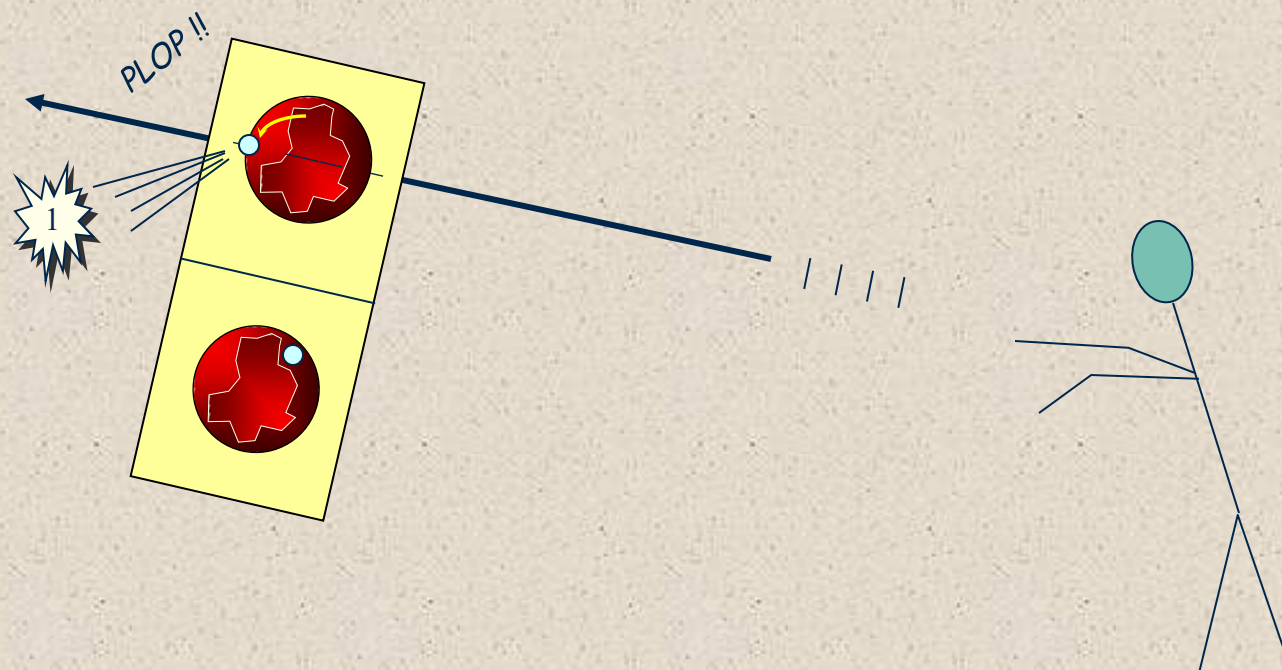
# Reading a 2-qubit register [2]



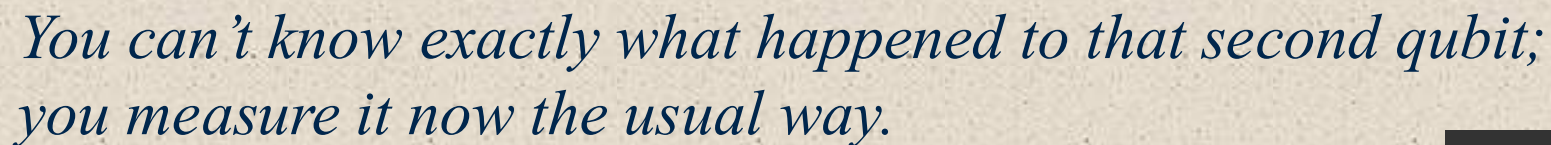
*Measure! The qubits become distinct now. The qubit you read jumps randomly to one of the two measurement poles.*

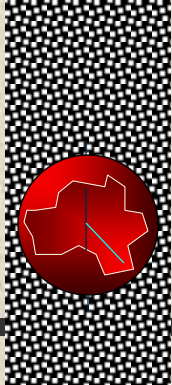


# Reading a 2-qubit register [3]



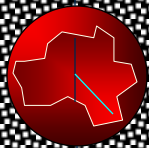
*If the random jump goes the other way, the second qubit can end up in a different state!*





# 3

## One Qubit

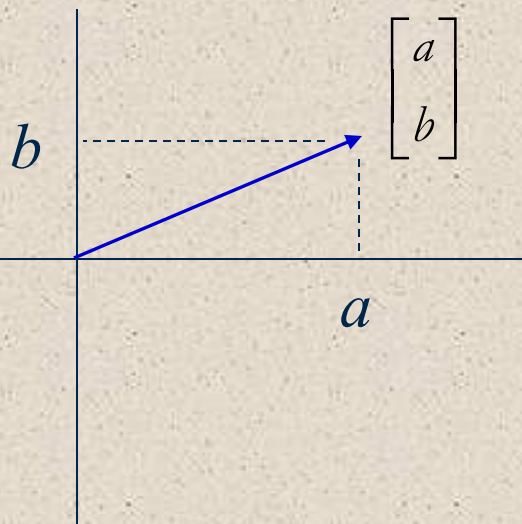


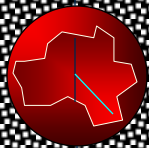
# Where do qubits come from?

They are based on **2-level quantum systems** modeled with **spinors**, which represent **quantum states**.

**Spinors** are *vectors* in two-dimensional complex Euclidean space ( $\mathbb{C}^2$ ).

This is  $\mathbb{C}^2$ , **not**  $\mathbb{R}^2$ ,  
so be careful  
interpreting this  
picture!





# State + State ??

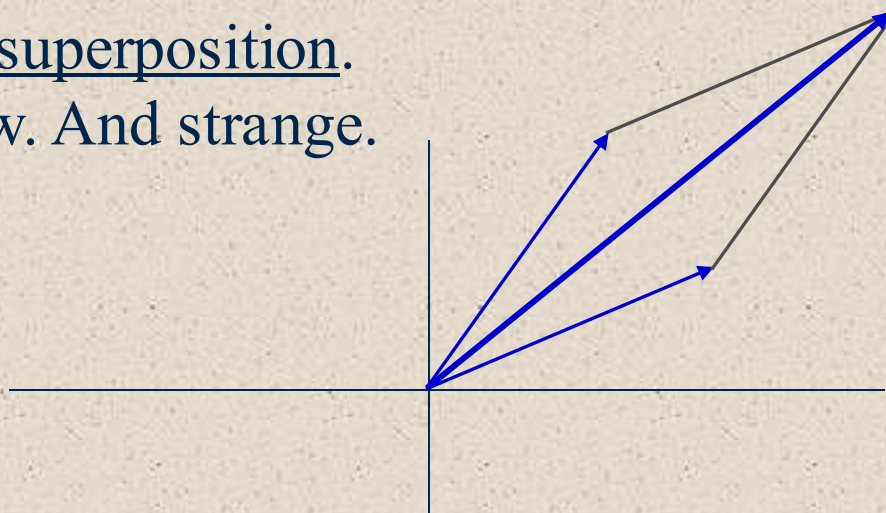
“Quantum states are represented by vectors”

= “The state space is a vector space”

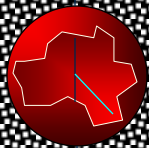
= “You can **add** states to get new states”

This is superposition.

It is new. And strange.

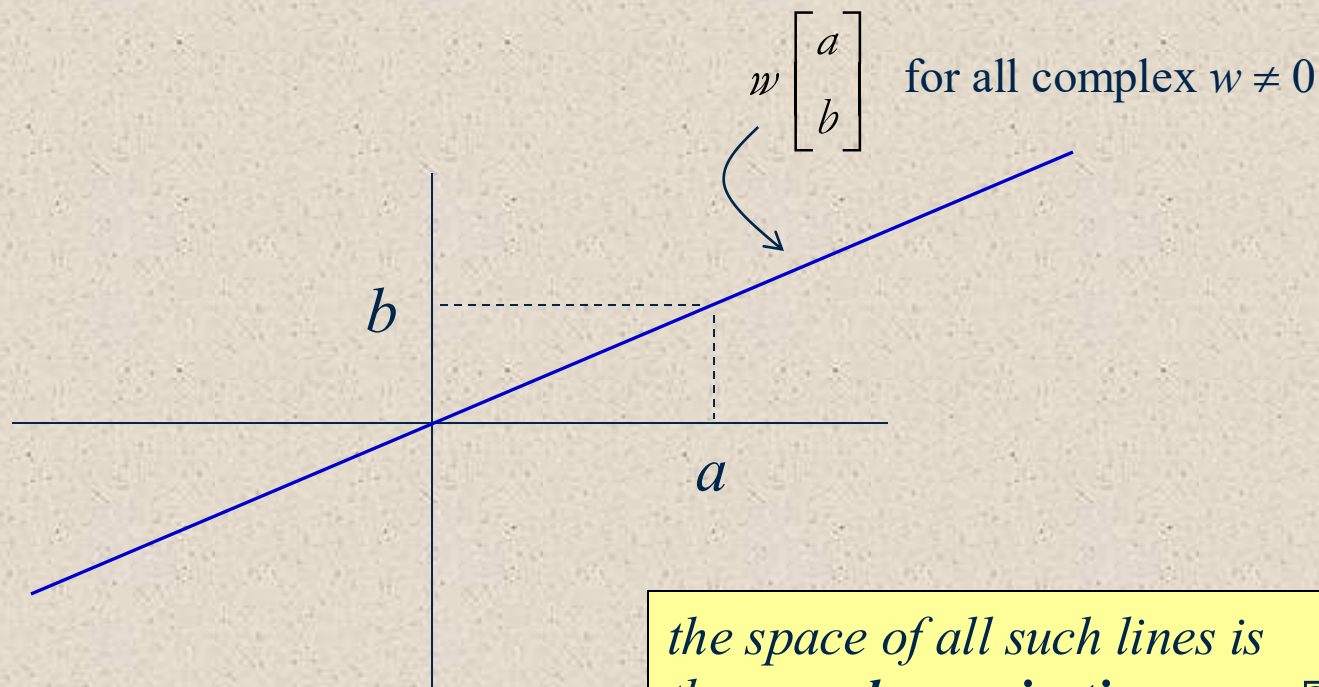


$q + q'$  is meaningless in automata theory  
 $(q,p) + (q', p')$  is meaningless in classical mechanics

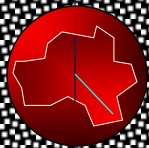


# Where do qubits come from?

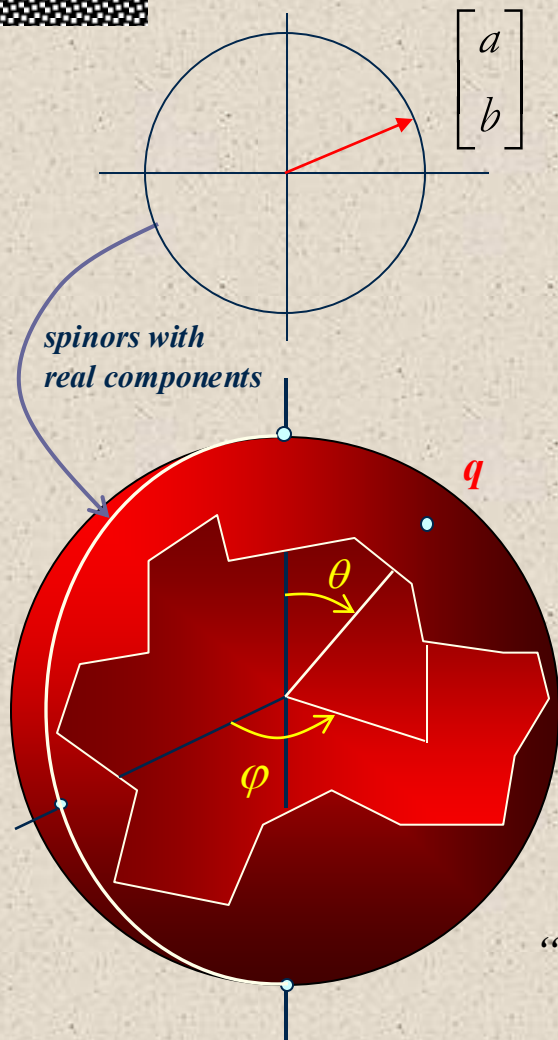
**Qubits** are *lines* in two-dimensional complex Euclidean space.



*the space of all such lines is  
the **complex projective space**  $\mathbb{CP}^1$*



# Spinor picture vs. qubit picture



Spinor picture.

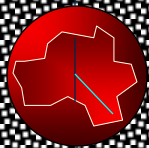
Incomplete *and* redundant.

But useful in calculation.

Qubits can be expressed as points on a sphere with coordinates  $(\theta, \varphi)$  using the relation

$$\begin{bmatrix} a \\ b \end{bmatrix} = w \begin{bmatrix} \cos \theta / 2 \\ e^{i\varphi} \sin \theta / 2 \end{bmatrix}$$

↑  
ignorable complex factor



# Analogy to representations in computer graphics

## Graphics

A vector in  $\mathbb{R}^4$   
 $\mathbf{x} = [w \ x \ y \ z]^T$   
 (“homogenous coordinates”).

*Good for  
calculation.*



*represents*

A point  $\mathbf{p}$  in real projective 3-space  $\mathbb{P}^3$ , pictured as Euclidean space plus points at infinity.

*Good for  
visualization.*

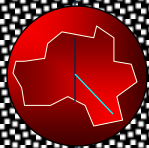
## Quantum computing

A vector in  $\mathbb{C}^2$ .  
 $\mathbf{q} = [q_0 \ q_1]^T$   
 (“spinor”).

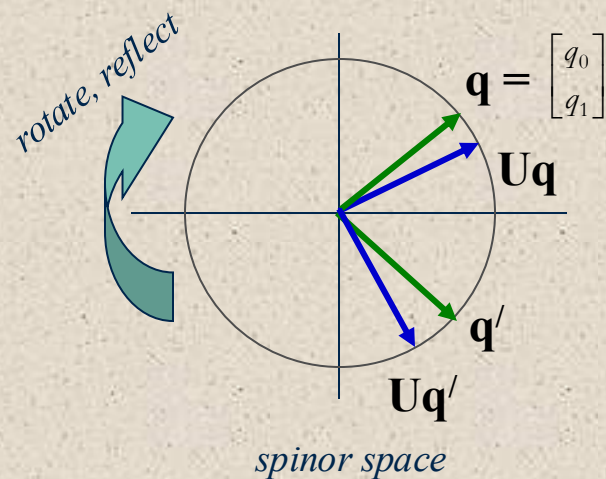


*represents*

A point (“qubit”) in complex projective 1-space  $\mathbb{P}^1$ , pictured as the surface of a sphere.



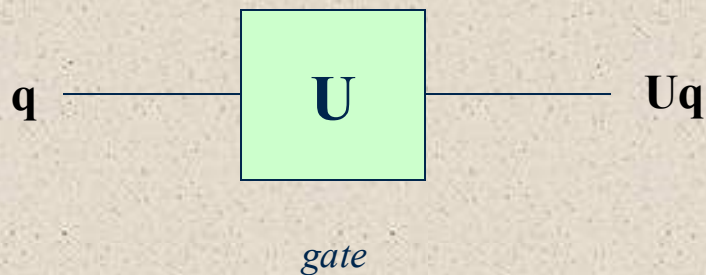
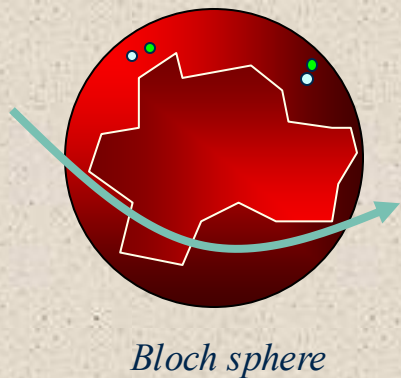
# States change 2 ways <sup>[1]</sup>

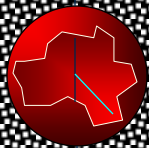


## 1. Unitary evolution

$$U = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \quad (U^* = U^{-1})$$

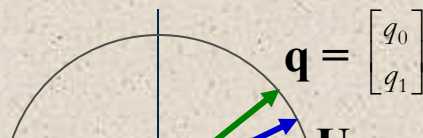
*continuous, deterministic, reversible*





# States change 2 ways <sup>[1]</sup>

rotate, reflect



## 1 Unitary evolution

**Schrödinger's Equation:**

$$d\mathbf{q}/dt = -i \mathbf{H} \mathbf{q}$$

This gives  $\mathbf{q}(t) = \mathbf{U}(t)\mathbf{q}(0)$ , where  $\mathbf{U}(t) = \exp -i\mathbf{H}t$ .



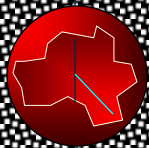
*Bloch sphere*

$\mathbf{q}$

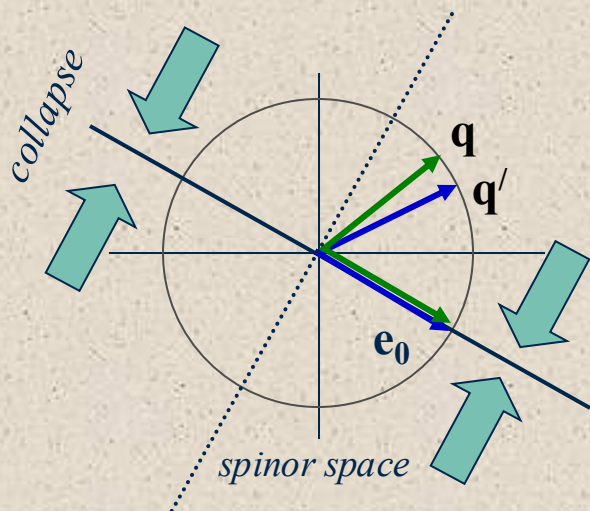
$\mathbf{U}$

$\mathbf{U}\mathbf{q}$

*gate*



# States change 2 ways [2.0]



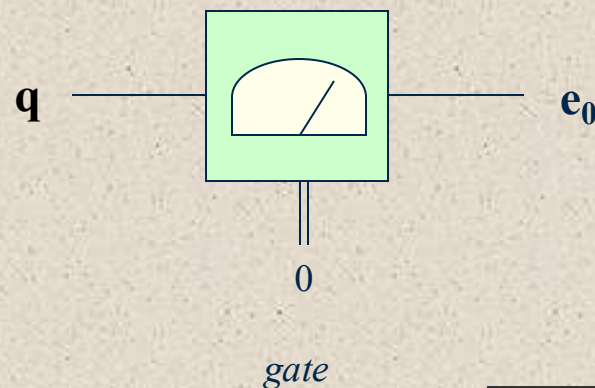
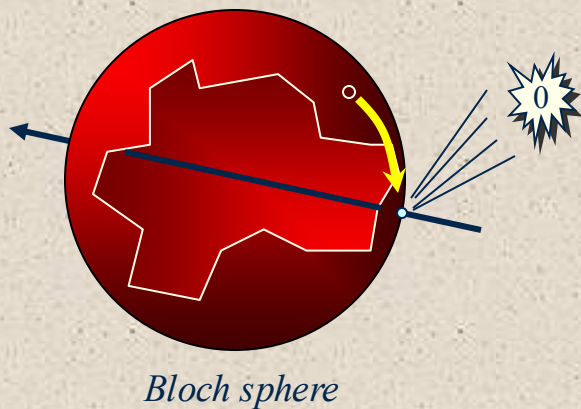
## 2. Measurement projection

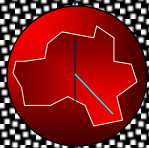
*discrete, stochastic, irreversible*

With probability  $|\mathbf{q} \cdot \mathbf{e}_0|^2$   
jump to  $\mathbf{e}_0$  and emit bit 0

else

jump to  $\mathbf{e}_1$  and emit bit 1.



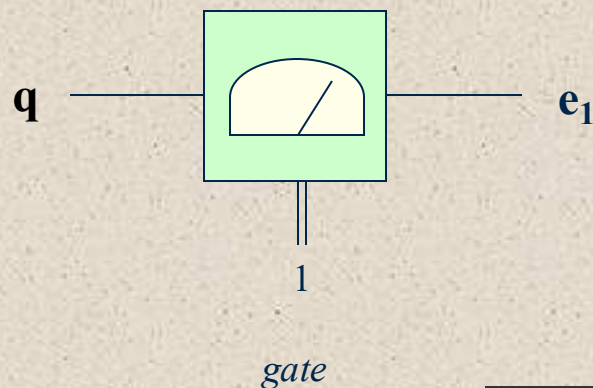
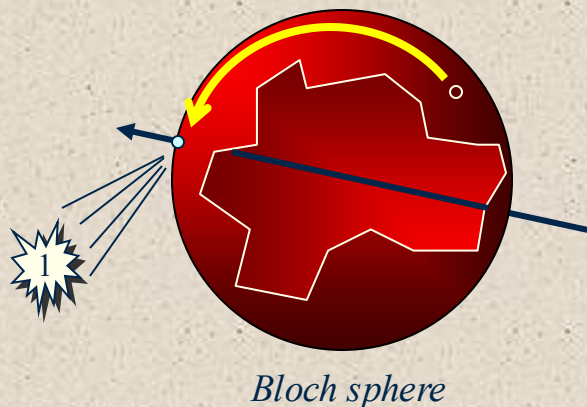
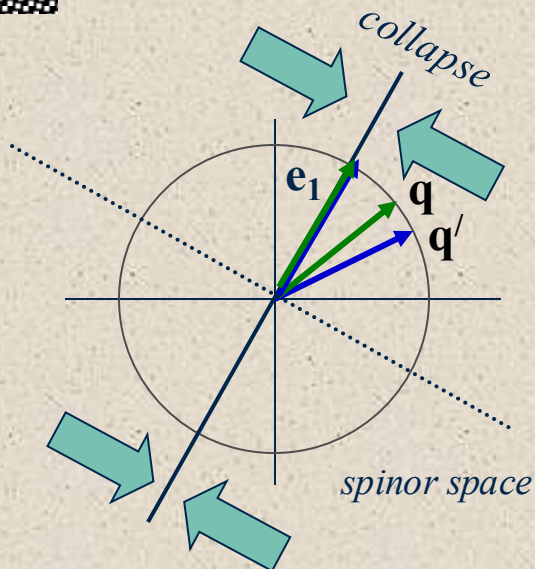


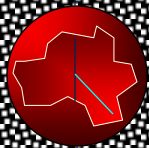
# States change 2 ways [2.1]

## 2. Measurement projection

*discrete, stochastic, irreversible*

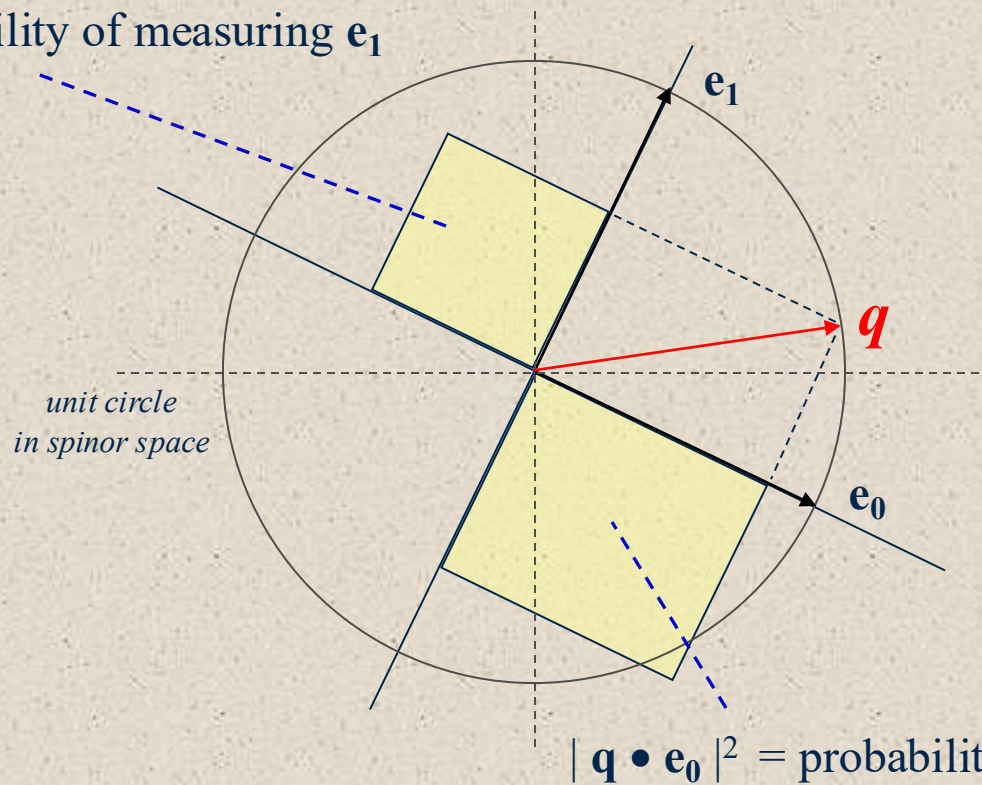
With probability  $|\mathbf{q} \cdot \mathbf{e}_0|^2$   
jump to  $\mathbf{e}_0$  and emit bit 0  
else  
jump to  $\mathbf{e}_1$  and emit bit 1.

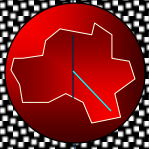




# Probabilities as areas

$|\mathbf{q} \cdot \mathbf{e}_1|^2 = \text{probability of measuring } \mathbf{e}_1$





# Probabilistic bits

A **probabilistic** (“mixed”) **bit** is one that is in different states with different probabilities.

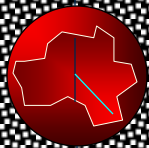
$$S1 = \left\{ \begin{array}{l} 0 @ 50\% ; \quad 1 @ 50\% \end{array} \right\}$$

$$S2 = \left\{ \begin{array}{l} 0 @ 15\% ; \quad 1 @ 85\% \end{array} \right\}$$



*simplex*

Representable as single real numbers  $p \in [0,1]$ .



# Probabilistic qubits

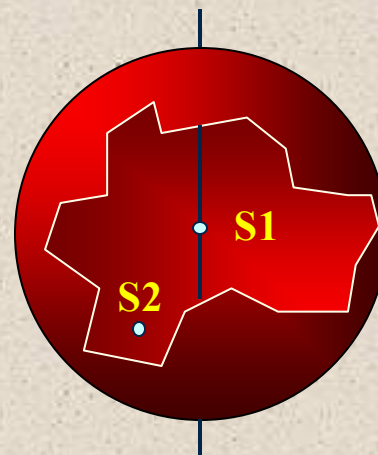
A **probabilistic** (“mixed”) **qubit** is one that is in different states with different probabilities.

$$S1 = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} @ 50\% ; \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} @ 50\% \right\}$$

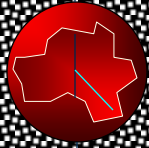
Mixed states are best represented as *matrices*.  
(No time to introduce that here, unfortunately.)

$$S2 = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} @ 70\% ; \quad \begin{bmatrix} 2 \\ 5 \end{bmatrix} @ 30\% \right\}$$

They can also be visualized as points *inside* the Bloch sphere.

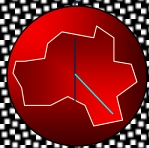


Non-mixed  
states are  
called “pure”.



# 4

## Several Qubits



# More bits

---

A 2-bit string is a string of two bits:

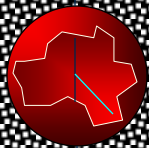


A 3-bit string is a string of three bits:



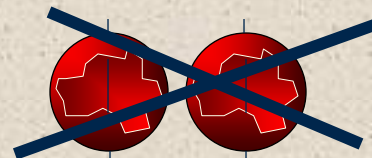
*etc.*

*No big deal.*

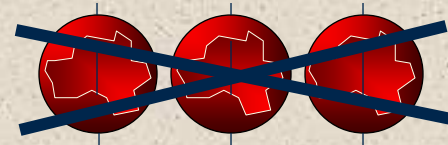


# More qubits

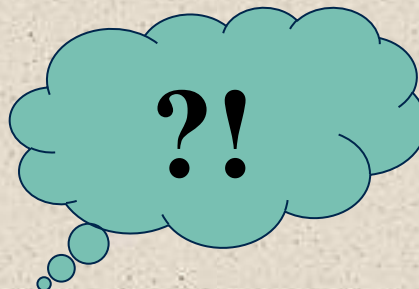
A 2-qubit string is **NOT** a string of two qubits:

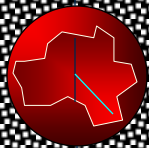


A 3-qubit string is **NOT** a string of three qubits:



*etc.*





# Building bit strings

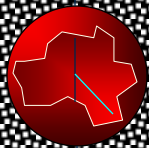
---

Let  $B = \{0,1\}$ .

A bit string  $\mathbf{b}$  is an element of  $\underline{B \times B \times \dots \times B} = B^n$



**cartesian product**



# Building qubit strings

Let  $B = \{0,1\}$ .

A bit string  $\mathbf{b}$  is an element of  $\underline{B \times B \times \dots \times B} = B^n$



**cartesian product**

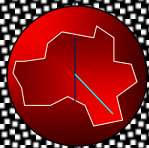
Let  $\mathbb{F}^2 = \{ \text{spinors} \}$

A qubit string  $\mathbf{s}$  is an element of  $\underline{\mathbb{F}^2 \otimes \mathbb{F}^2 \otimes \dots \otimes \mathbb{F}^2} = (\mathbb{F}^2)^{\otimes n}$



**tensor product**

(Confusing  $A \times B$  with  $A \otimes B$  is like confusing  $2+5$  with  $2 \times 5$  !)



# Direct sums and tensor products *of vectors*

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \times \begin{bmatrix} 1 \\ 10 \\ 100 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \\ 10 \\ 100 \end{bmatrix}$$

*Also called a direct  
sum “ $\oplus$ ”*

$$\mathbb{R}^3 \times \mathbb{R}^3 \simeq \mathbb{R}^6$$

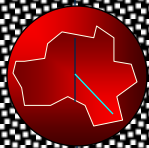
**Cartesian product**

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 10 \\ 100 \end{bmatrix} = \begin{bmatrix} 2 \\ 20 \\ 200 \\ 3 \\ 30 \\ 300 \\ 4 \\ 40 \\ 400 \end{bmatrix}$$

*Also called a  
Kronecker product*

$$\mathbb{R}^3 \otimes \mathbb{R}^3 \simeq \mathbb{R}^9$$

**tensor product**



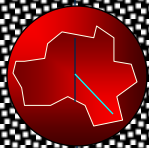
# The general 2-qubit string

Define...

$$\mathbf{e}_{00} = \mathbf{e}_0 \otimes \mathbf{e}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{e}_{01} = \mathbf{e}_0 \otimes \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \text{ etc.}$$

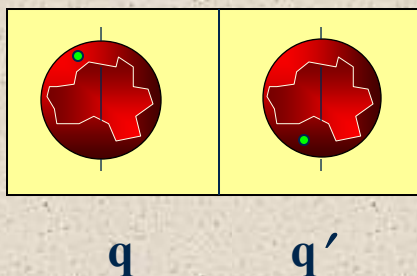
A general 2-qubit string is represented this way:

$$\mathbf{q} = h_{00} \mathbf{e}_{00} + h_{01} \mathbf{e}_{01} + h_{10} \mathbf{e}_{10} + h_{11} \mathbf{e}_{11} = \begin{bmatrix} h_{00} \\ h_{01} \\ h_{10} \\ h_{11} \end{bmatrix}$$



# 2 qubits: a special case

The simplest case is when 2 qubits are in a **product state**.



$$q \otimes q'$$

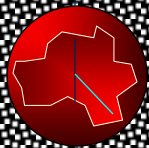
That is, the state looks like this product of spinors:

$$q \otimes q' = (\underline{ae_0 + be_1}) \otimes (\underline{ce_0 + de_1}) = \underline{ac e_{00} + ad e_{01} + bc e_{10} + bd e_{11}}$$

That is:

$$\begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac \\ ad \\ bc \\ bd \end{bmatrix}$$

*In this special case,  
a 2-qubit string  
is a string of 2 qubits.*



# Examples: Product vs. entangled

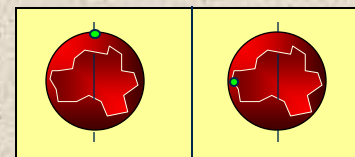
## The Simplest Case: A Product State

$$e_{00} + e_{01} = (1e_0 + 0e_1) \otimes (1e_0 + 1e_1)$$

so it is a product of 2 qubits.

You could treat it as a “string” of single qubits.

$$e_{00} + e_{01}$$



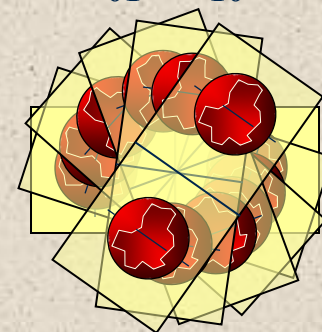
## The General Case : An **Entangled** State

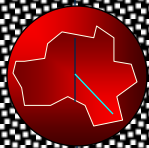
$$e_{01} + e_{10} \neq (ae_0 + be_1) \otimes (ce_0 + de_1) \text{ for any } a, b, c, d.$$

It is not equal to  $q \otimes q'$  for any  $q, q'$

You cannot treat it as a “string” of single qubits.

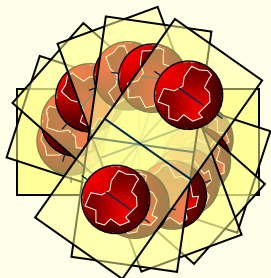
$$e_{01} + e_{10}$$





# Example

Let's read a 2-qubit "register" in the following (entangled) state:

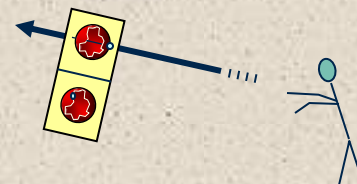


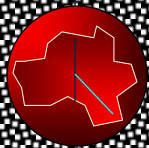
$$\left(\frac{1}{\sqrt{30}}\right) e_{00} + \left(\frac{2}{\sqrt{30}}\right) e_{01} + \left(\frac{3}{\sqrt{30}}\right) e_{10} + \left(\frac{4}{\sqrt{30}}\right) e_{11}$$

*Let's read out the first qubit in the  $e_0, e_1$  basis:*

We get **0** with probability  $|1/\sqrt{30}|^2 + |2/\sqrt{30}|^2 = \mathbf{1/6}$ ,  
and the second qubit jumps to state  $\left(\frac{1}{\sqrt{5}}\right) e_0 + \left(\frac{2}{\sqrt{5}}\right) e_1$ .

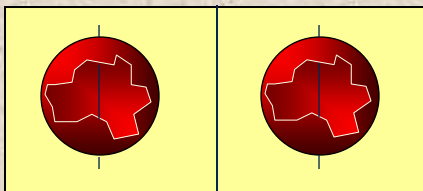
We get **1** with probability  $|3/\sqrt{30}|^2 + |4/\sqrt{30}|^2 = \mathbf{5/6}$ ,  
and the second qubit jumps to state  $\left(\frac{3}{5}\right) e_0 + \left(\frac{4}{5}\right) e_1$ .



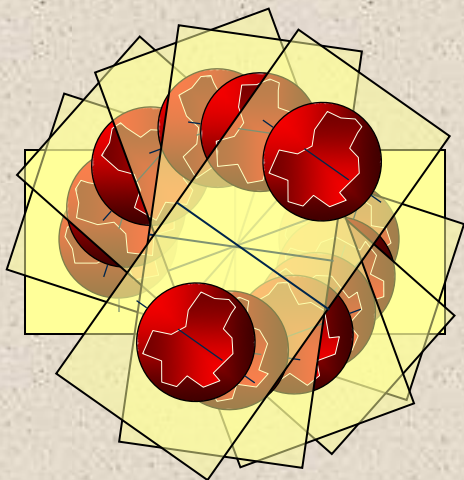


## 2 qubits: summary

---

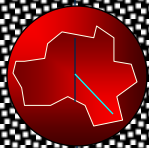


When the qubits are *not* entangled, reading one qubit does not affect the state of the other qubit.



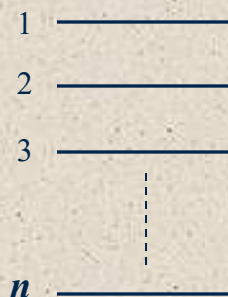
When the qubits are *are* entangled, reading one qubit *does* affect the state of the other qubit.

*Very strange, but entanglement is a valuable resource in quantum computing.*



# An $n$ -qubit string

An  $n$ -qubit string is represented by a vector in  $2^n$ -dimensional space.



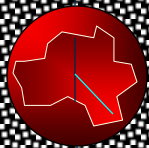
$$\mathbf{q} = \sum_{i=0}^{2^n-1} h_i \mathbf{e}_i$$

$$= h_0 \mathbf{e}_{00..0} + h_1 \mathbf{e}_{00..1} + \dots + h_{(2^n-1)} \mathbf{e}_{111..1}$$

*Hmmm...*

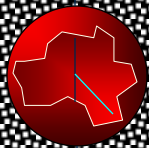
This uses only  $O(n)$  physical resources.

Can we use all this "free extra space" to solve problems? Yes...



# 5

## Some quantum gates, circuits and algorithms.



# Dirac notation: Quick & Sloppy

$$|0\rangle = \mathbf{e}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle = \mathbf{e}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

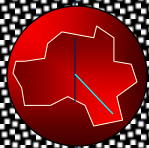
$$|0110\rangle = |0\rangle|1\rangle|1\rangle|0\rangle = \mathbf{e}_{0110} = \mathbf{e}_0 \otimes \mathbf{e}_1 \otimes \mathbf{e}_1 \otimes \mathbf{e}_0 =$$

and so on.

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$



slot 6  
(slot 0110)



# Gate notation

---

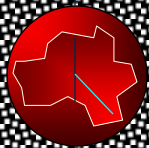


qubit

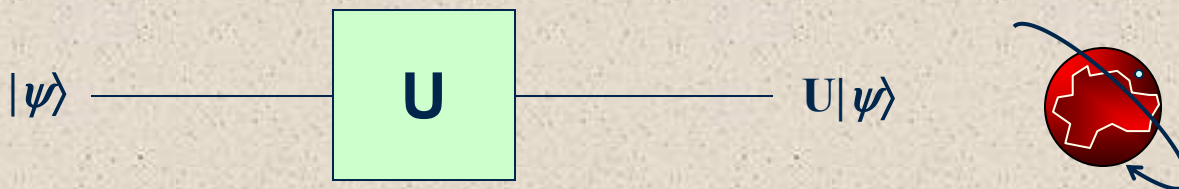


bit

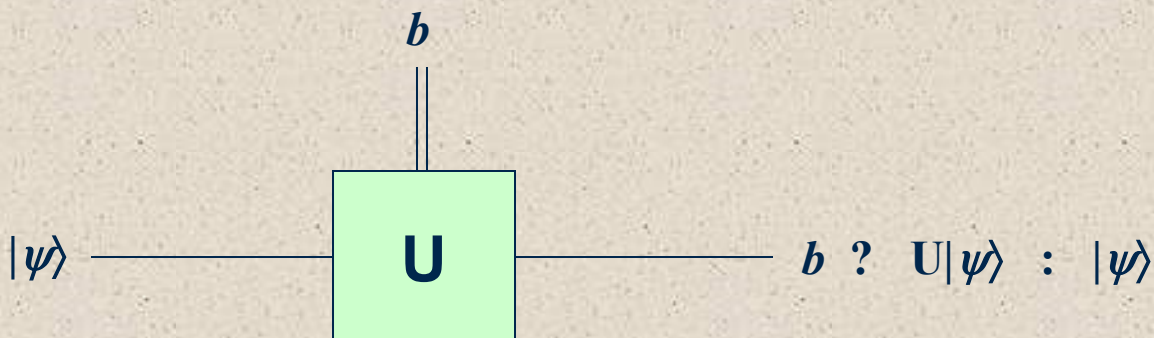


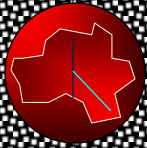


# Gate notation



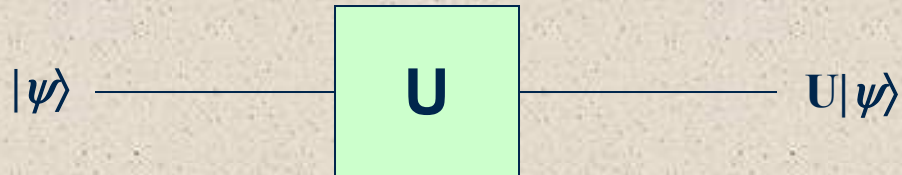
Turning a gate  
on/off with  
an input bit



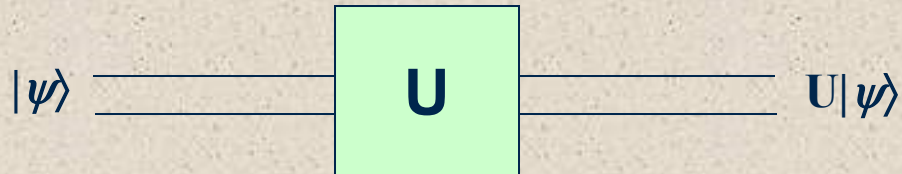


# Gate notation

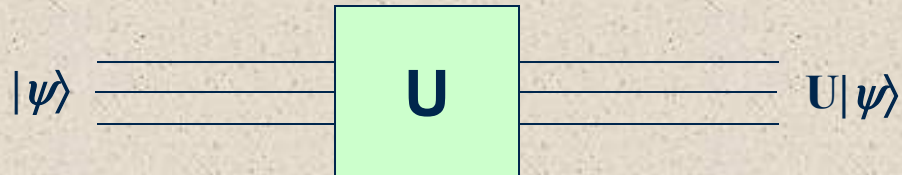
$U$  is a  **$2 \times 2$**   
complex unitary  
matrix

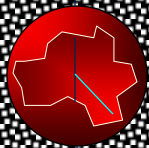


$U$  is a  **$4 \times 4$**   
complex unitary  
matrix

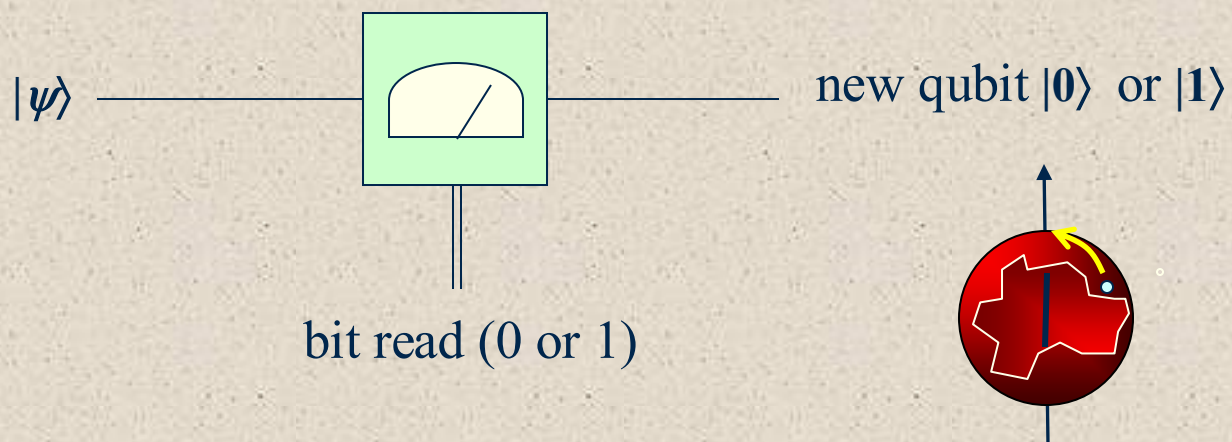


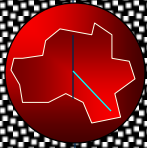
$U$  is a  **$8 \times 8$**   
complex unitary  
matrix





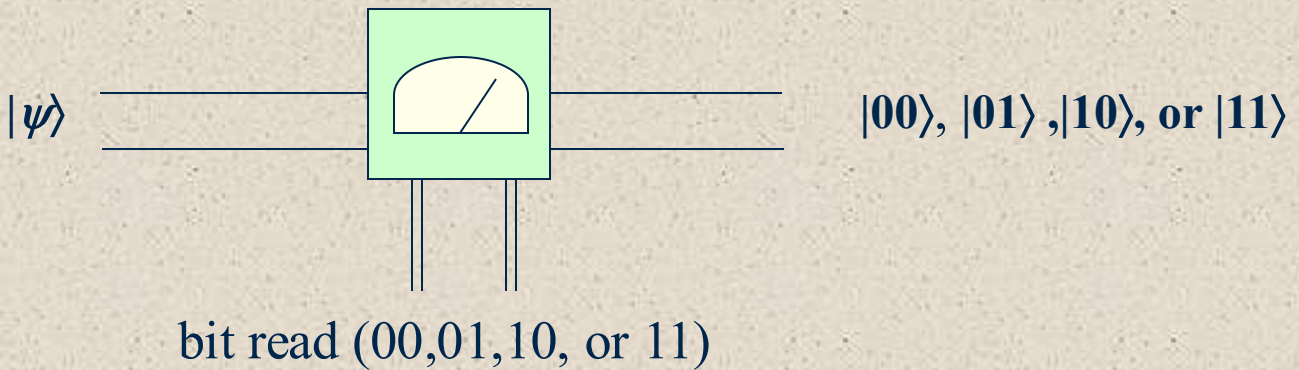
# Gate notation

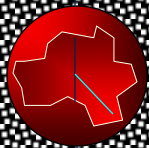




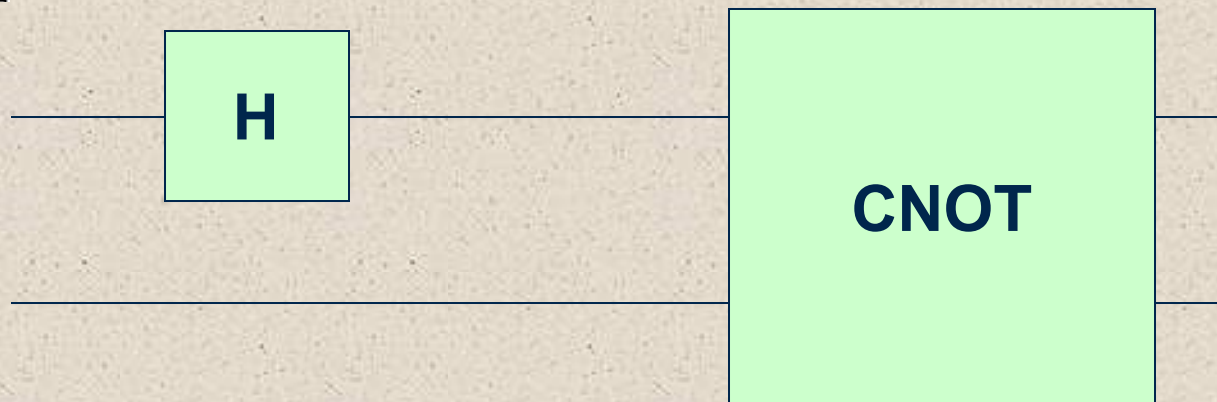
# Gate notation

*Measuring 2-qubits simultaneously.*





# A quantum circuit for creating entanglement <sup>(0)</sup>



$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

## Hadamard transform

a reflection in the first qubit's space

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

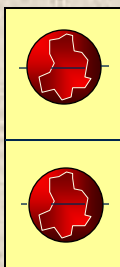
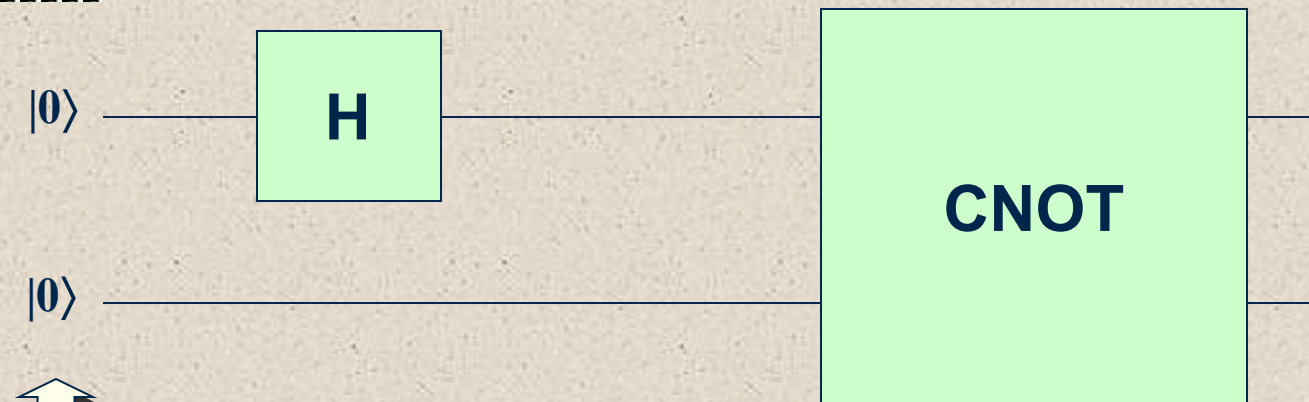
## Controlled-NOT

If 1st qubit is  $|0\rangle$ , do nothing.

If 1st qubit is  $|1\rangle$ , flip 2nd qubit  $|0\rangle \leftrightarrow |1\rangle$

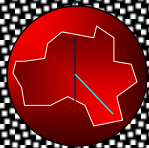


# A quantum circuit for creating entanglement (1)

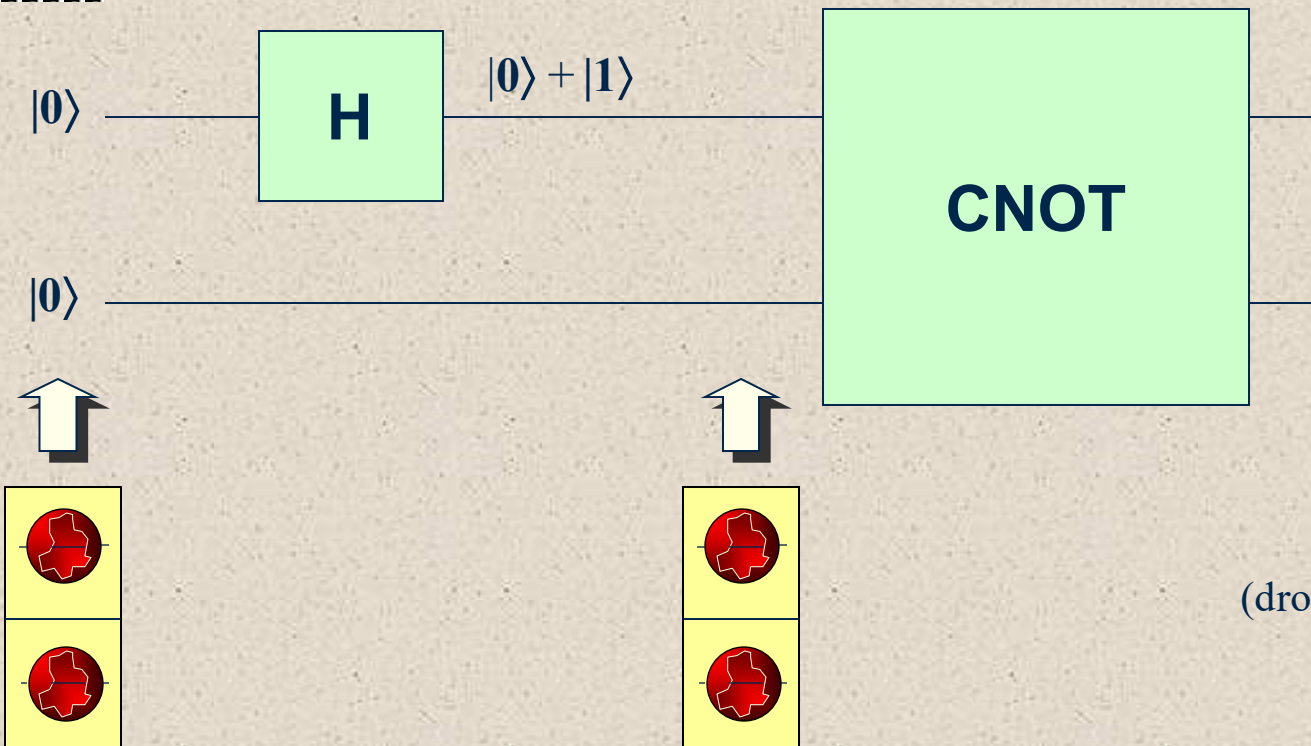


not entangled

$|0\rangle|0\rangle$



## A quantum circuit for creating entanglement (2)

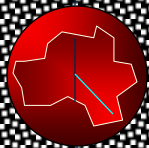


(dropping the  $2^{-1/2}$  coefficients  
for readability)

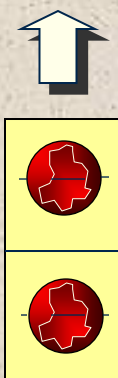
not entangled  
 $|0\rangle|0\rangle$

*still* not entangled

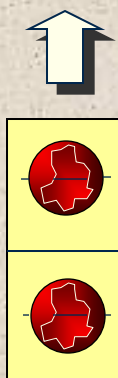
$$(|0\rangle + |1\rangle) |0\rangle = |0\rangle|0\rangle + |1\rangle|0\rangle$$



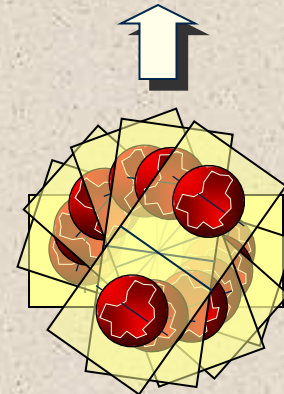
# A quantum circuit for creating entanglement <sup>(3)</sup>



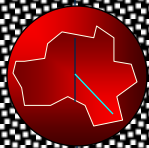
not entangled  
 $|0\rangle|0\rangle$



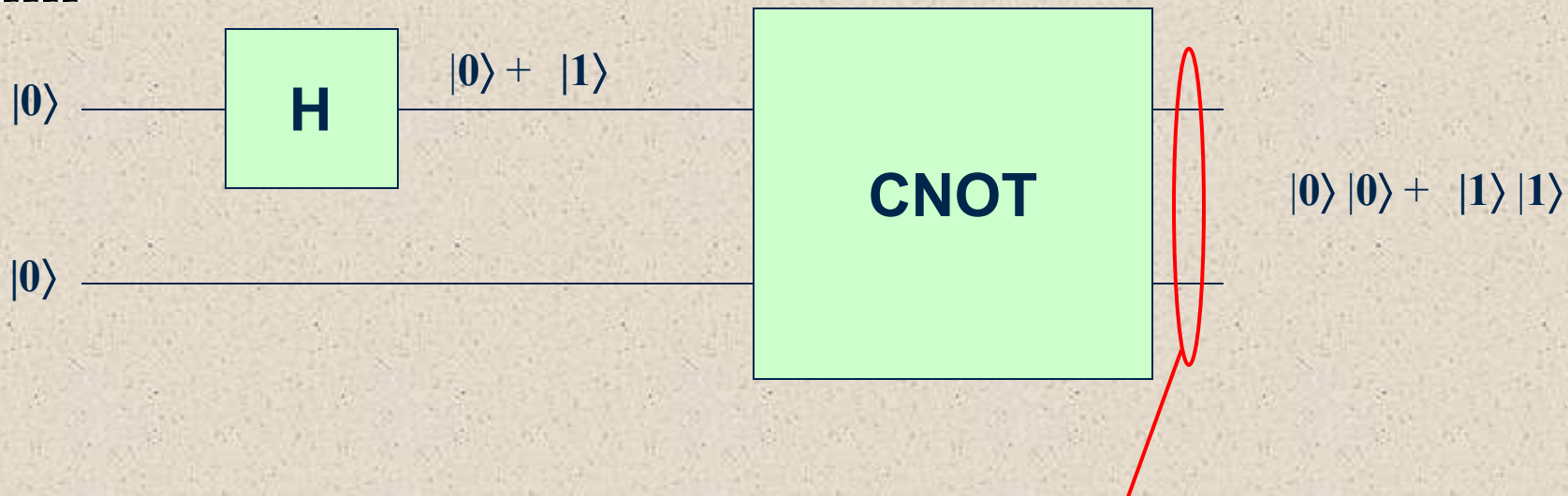
still not entangled  
 $|0\rangle|0\rangle + |1\rangle|0\rangle$



**entangled !**  
 $|0\rangle|0\rangle + |1\rangle|1\rangle$

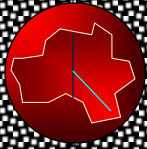


## A quantum circuit for creating entanglement (4)

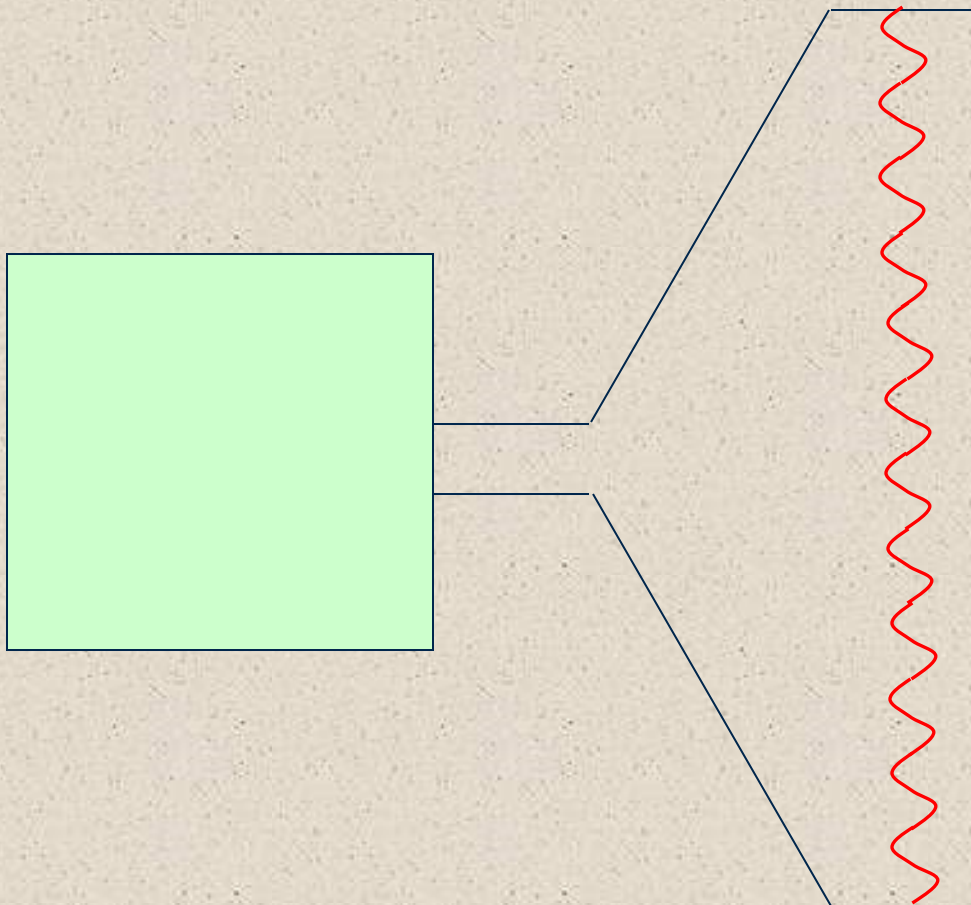


**Unlike ordinary combinational logic circuits,  
two nearby wires cannot be assumed to be independent!**

**You must treat them *holistically*.**



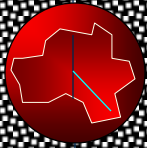
# *Spukhafte Fernwirkungen*



Even if these 2 qubits are sent  
*very* far apart,  
reading one qubit  
**instantaneously** affects  
the other qubit.

(Cannot use this for communication,  
since observing the differences  
requires bit transmission.)





# No “Fanout”

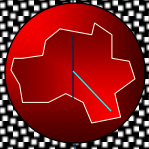
$|\psi\rangle$

$|\psi\rangle$

NO

$|\psi\rangle$

time



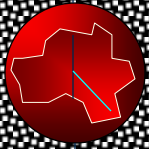
# Quantum teleportation *is* possible

$|\psi\rangle$

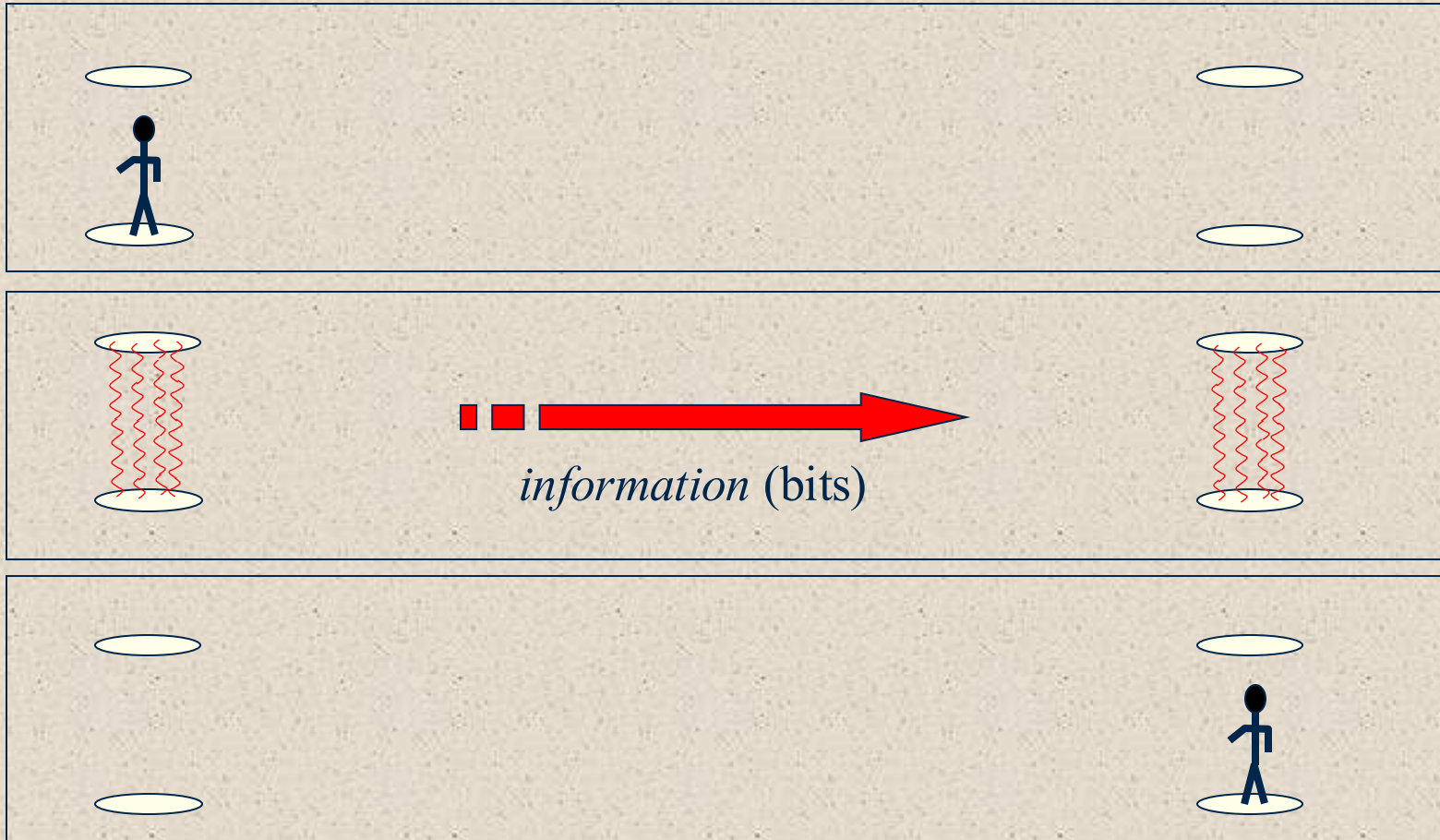
?

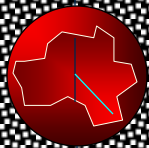
$|\psi\rangle$

time

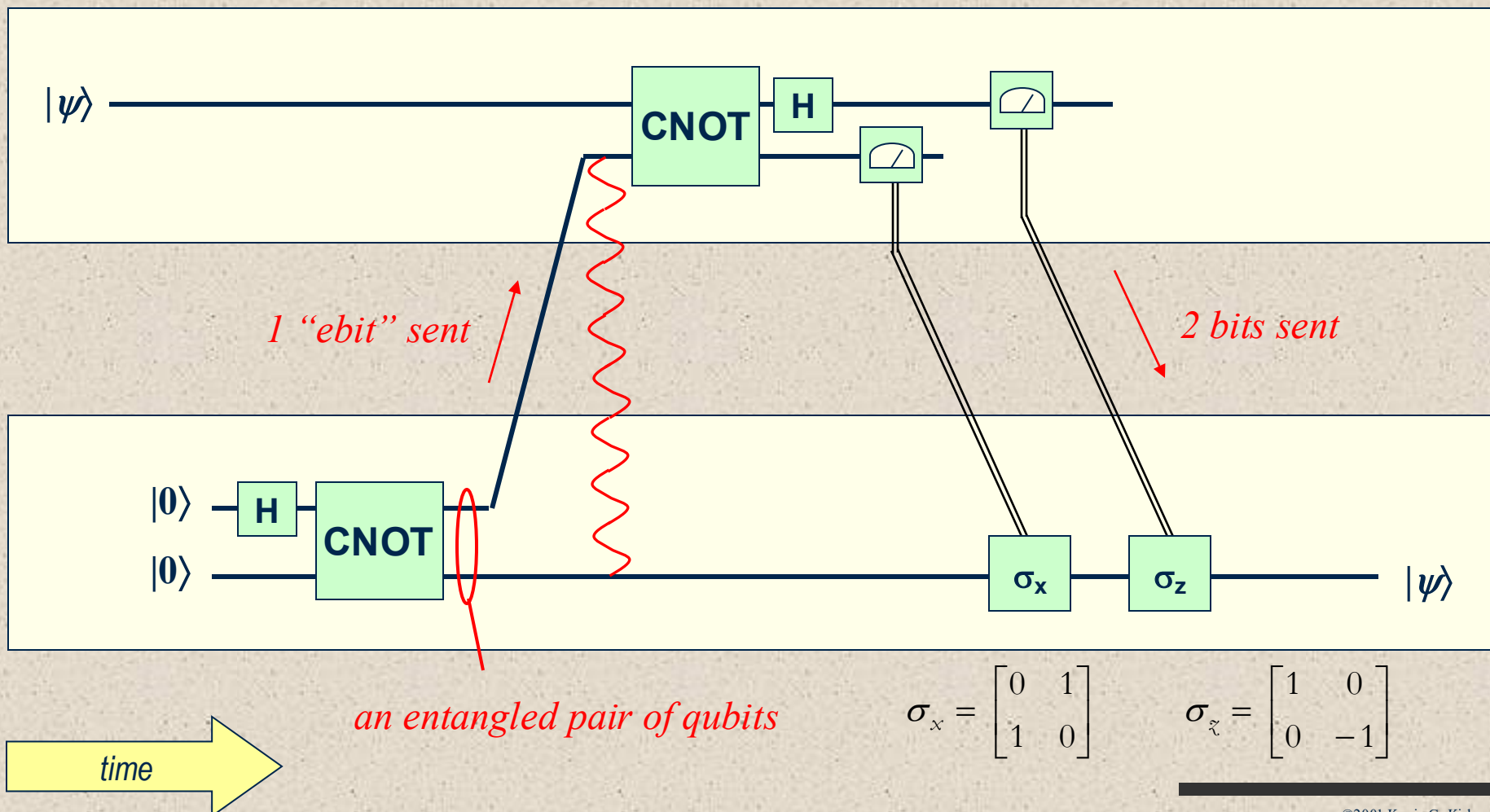


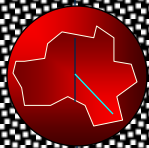
# Teleportation



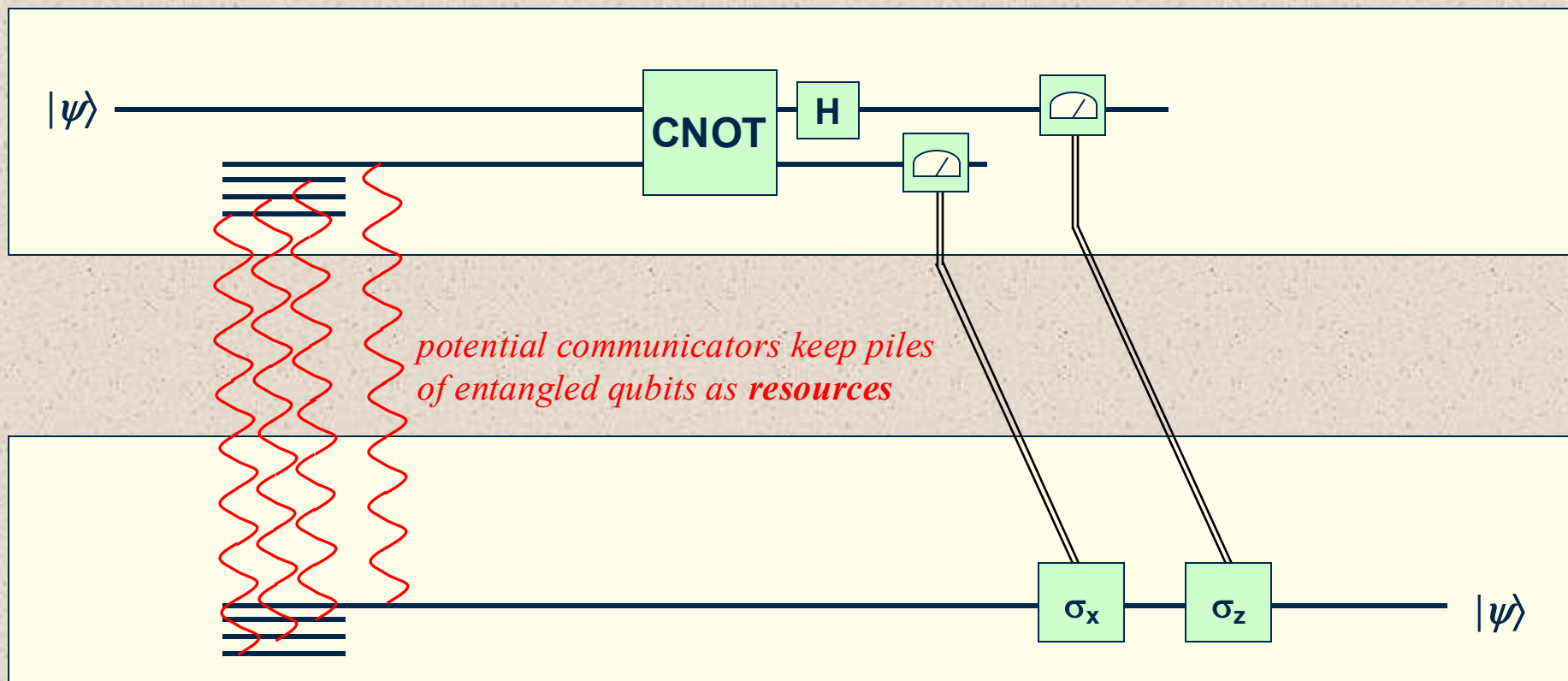


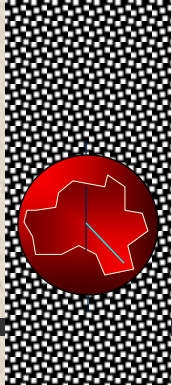
# Quantum teleportation circuit





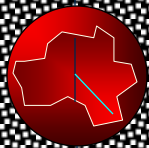
# Entanglement as a resource





# 6

## Quantum database search

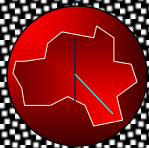


# Unstructured Search <sup>(1)</sup>

---

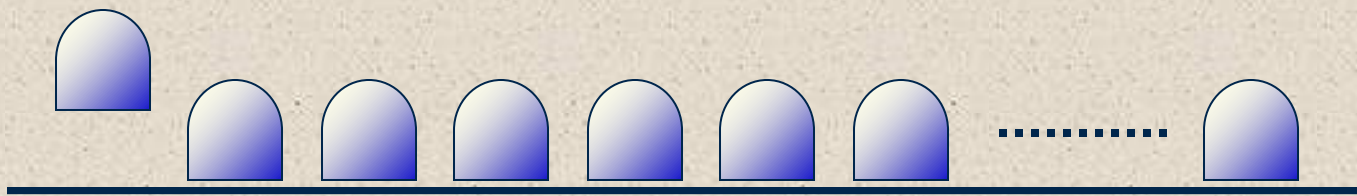


Where is it?

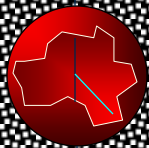


# Unstructured Search <sup>(2)</sup>

---

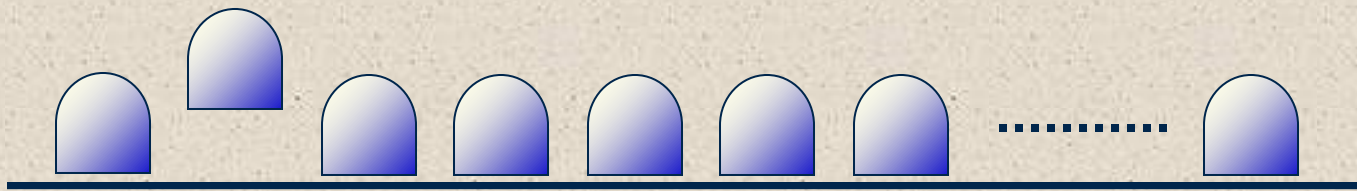


Here?

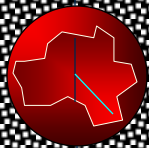


# Unstructured Search <sup>(3)</sup>

---



Here?

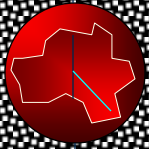


# Unstructured Search <sup>(4)</sup>

---



Here?

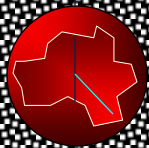


# Unstructured Search <sup>(5)</sup>

---

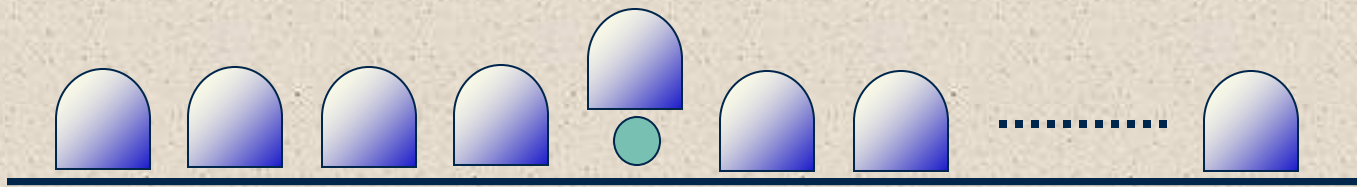


Here?

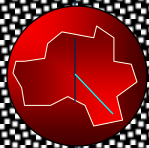


# Unstructured Search <sup>(6)</sup>

---

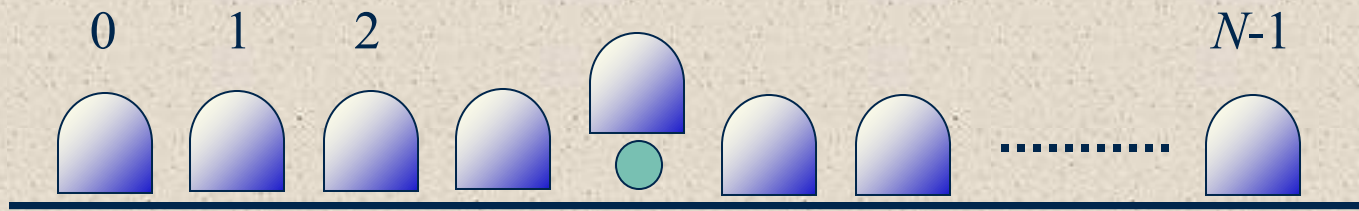


Found it.



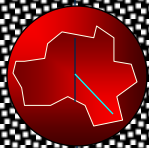
# Unstructured Search <sup>(7)</sup>

---

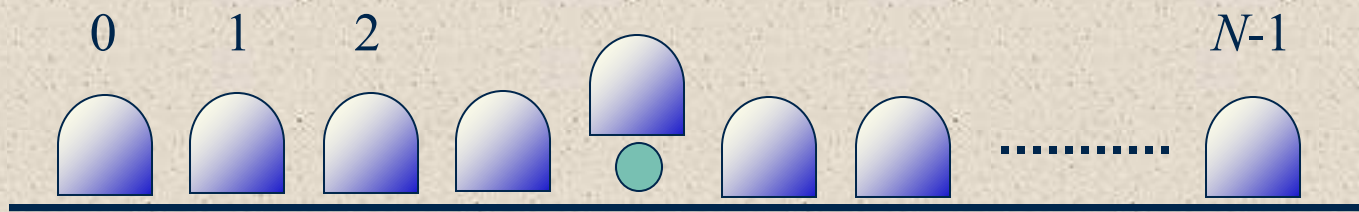


Found it.

Took  $O(N)$  peeks.



# Unstructured Search <sup>(7)</sup>

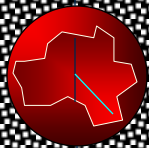


Found it.

Took  $O(N)$  peeks.

**Quantum algorithm:  $O(\sqrt{N})$  peeks.**





# The Setup

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Suppose we are searching  $N=2^n$  items.

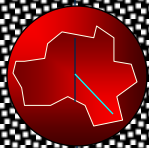
Use  $n$ -qubit strings to keep track of these items.

Example ( $N=8$ ):  $|000\rangle$   $|001\rangle$   $|010\rangle$   $|011\rangle$   $|100\rangle$   $|101\rangle$   $|110\rangle$   $|111\rangle$



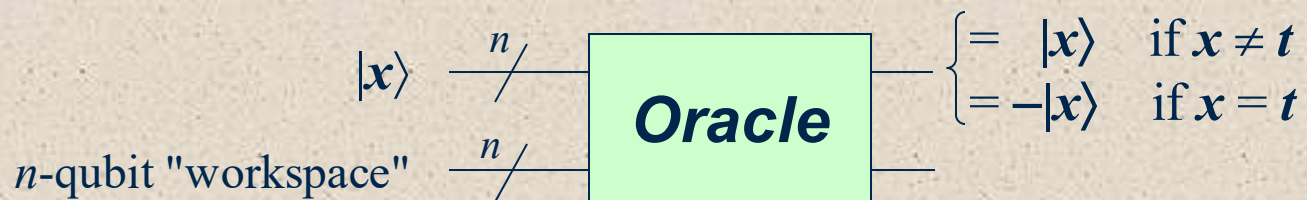
*One of these is the target; call it  $|t\rangle$  .*

We assume that given a state  $|x\rangle$  we could determine whether it was the target in  $O(1)$  time.



# The Oracle

We can build a quantum gate that recognizes the target and works like this.  
Let  $\mathbf{x}$  be in  $\{0,1\}^N$ .

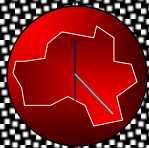


You can build this device **even though you don't know what  $t$  is**, just using a function that checks whether a given  $x$  is equal to  $t$ .

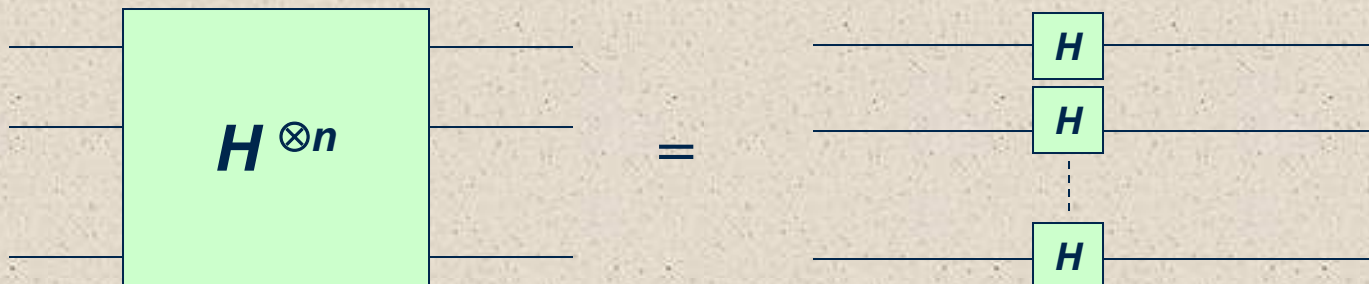
This is the way it is in many search problems.

**Testing** a solution is fast and easy.

**Finding** a solution is slow and hard, and requires the use of “**testing**” to do so.



# The most important quantum gate



Think of  $H$  as a "mix it up" gate:

$$H|0\rangle = [ |0\rangle + |1\rangle ] / 2^{1/2}$$

$$H|1\rangle = [ |0\rangle - |1\rangle ] / 2^{1/2}$$

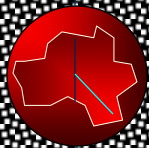
Think of  $H^{\otimes n}$  as an "all-possibilities" gate:

$$H^{\otimes 2} |00\rangle = [ |00\rangle + |01\rangle + |10\rangle + |11\rangle ] / 2$$

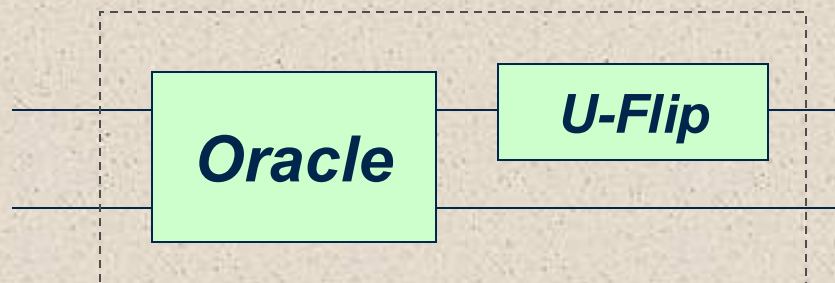
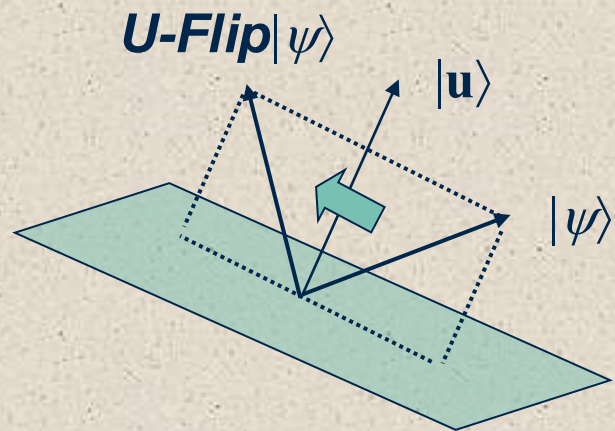
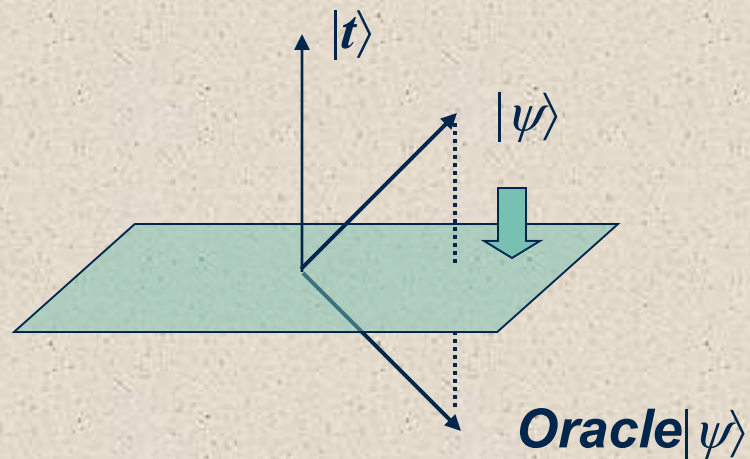
$$H^{\otimes 3} |000\rangle = [ |000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle ] / 2^{3/2}$$

etc.

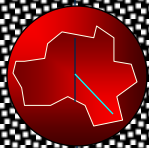
Define the *uniform vector*  $|u\rangle \equiv H^{\otimes n} |000\dots 0\rangle$



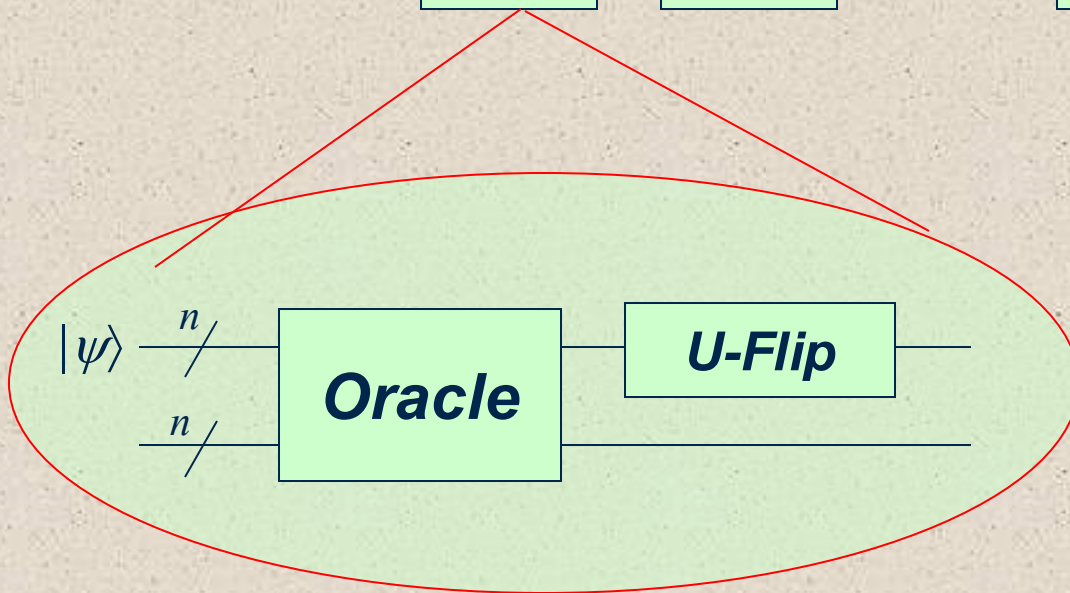
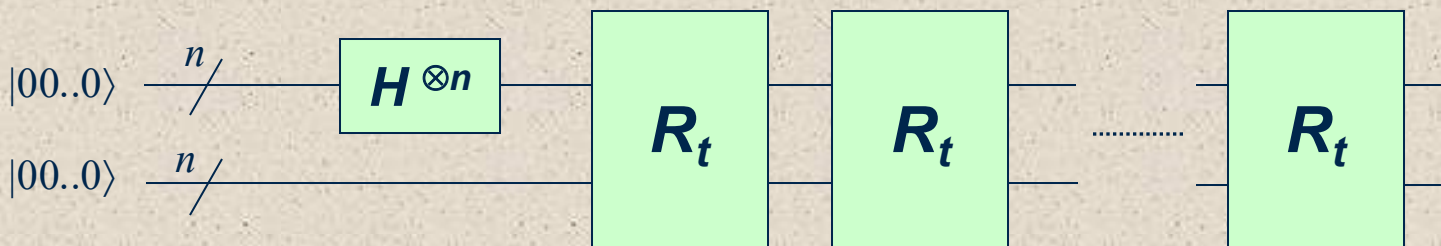
# The geometry of quantum search

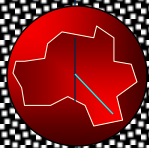


Combine them to get a **rotation**  $R_t = (U\text{-Flip})(Oracle)$

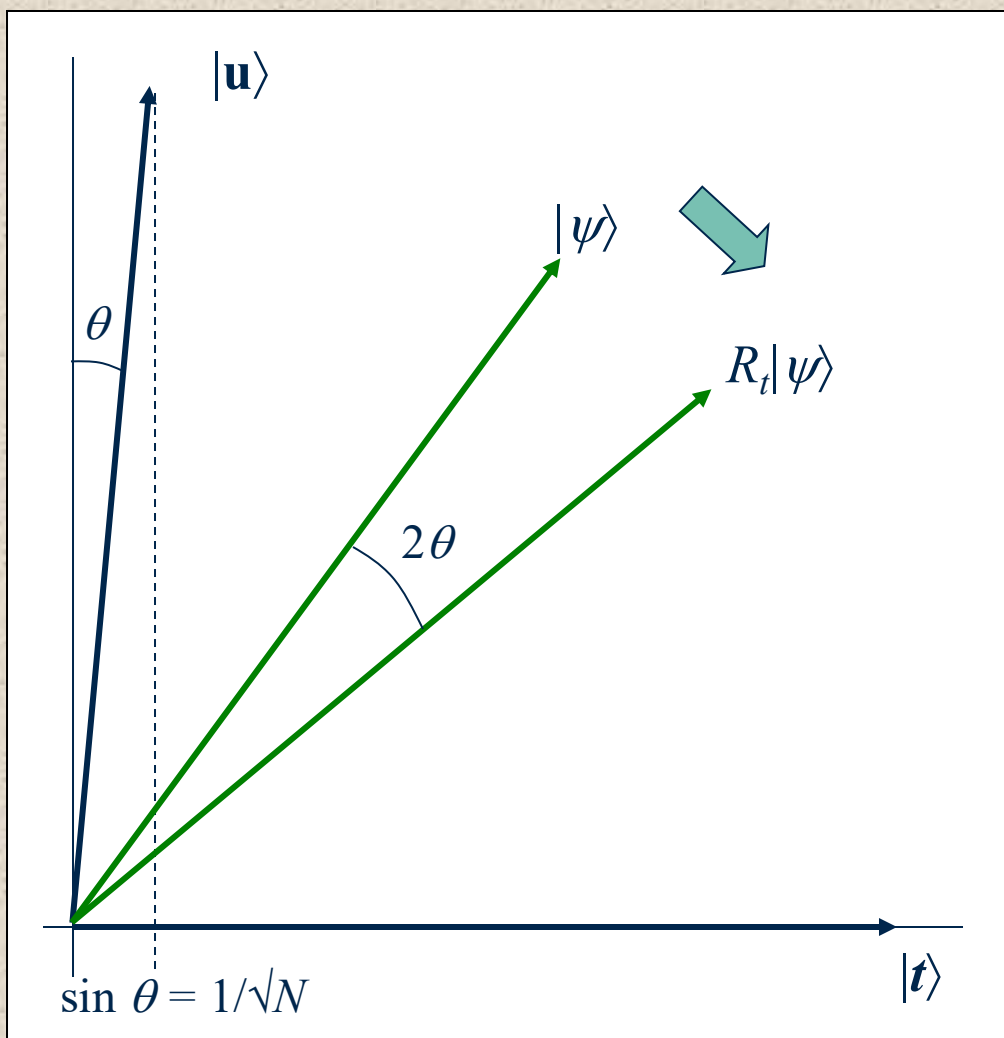


# The search algorithm



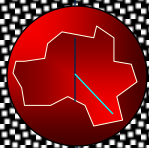


# The search rotation

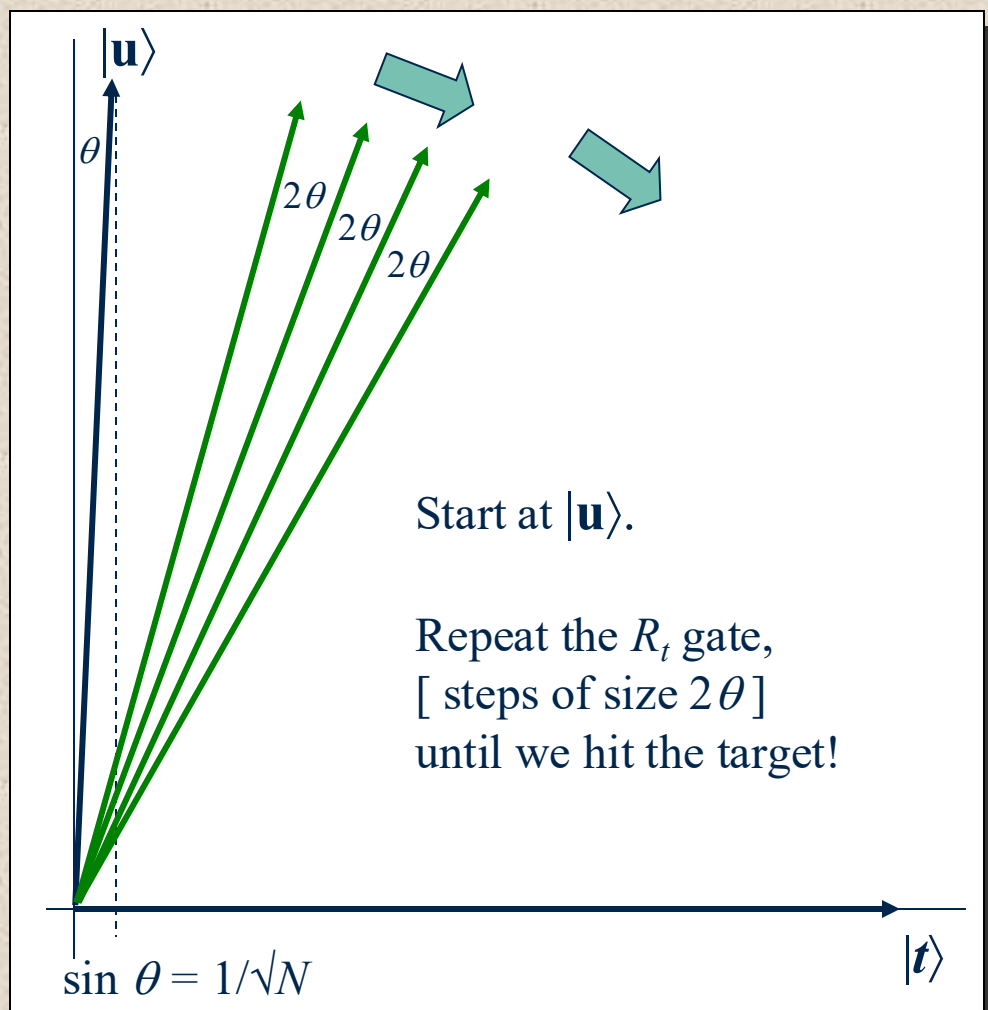


The  $|t\rangle, |u\rangle$  plane.  
(A tiny 2D subspace of a  
very high dimensional space!)

Note that we are  
interested in **large**  
databases:  
 $N$  will be very large!  
So  $\theta$  will be very small.



# Zooming in to the target....



**How many steps to find target?**

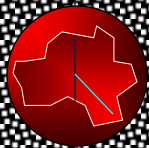
Let  $M$  be the number of steps  
until we are closest to the target.  
Then:

$$\pi/2 - \theta = M 2\theta$$

Use the small-angle approximation

$$\theta \approx \sin \theta = 1/\sqrt{N}$$

So we conclude  **$M = O(\sqrt{N})$** .  
An amazing and inspirational result  
(Grover, 1996).

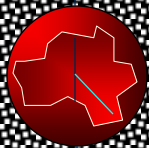


# Well, almost....

You might not hit the target **exactly**, but applying the **measurement collapse** gate projects it onto the target with extremely high probability (better as  $N$  gets larger).



As in any probabilistic algorithm, you can repeat it to increase the certainty of the result.



# Grover on his algorithm

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**CSC**

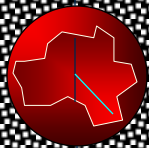
“Searching with Quantum Computers”

*Dr. Dobb's Journal* (April 2001).

**PHY**

“From Schrödinger's Equation to Quantum Search”

*American Journal of Physics* (July 2001).



# Is QC really faster?

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Unstructured search using QC:

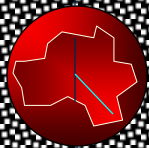
$$O(N) \rightarrow O(\sqrt{N}).$$

(This was later proven to be the best possible QC speedup in unstructured searching.)

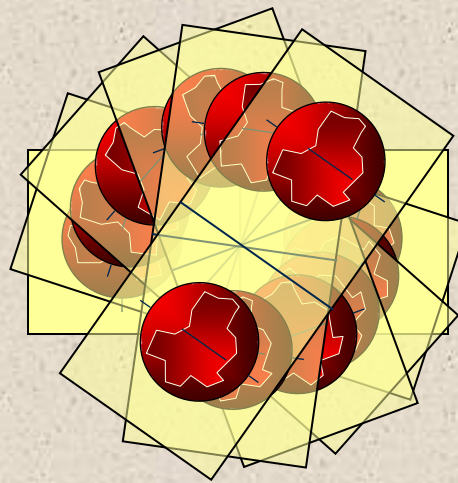
QC is capable of *exponential* speedup in factoring:

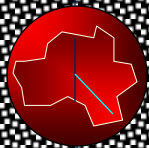
$$O(N) \rightarrow O(\log N).$$

It is believed (but unproven) that no classical algorithms can do that.



At present, QC is still **computer science**,  
not **computer technology**.





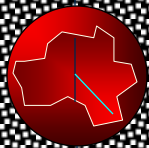
# Quantum Computing *Is* Different

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QC *is* a fundamental change in the foundations of computer science.

It gives us...

- a new theory of computation and complexity
- faster algorithms (not just faster implementations)
- a new understanding of information in nature



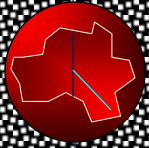
# Studying QC

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Most QC researchers are in in **physics**.

More **computer scientists** should study the field.

The most important prerequisite is  
**linear algebra.**

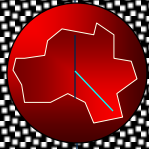


# Simple Samples from the Frontier

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“On the correct quantization of genetic algorithms”

“Quantum entropies and adaptability theory”



# Conclusions

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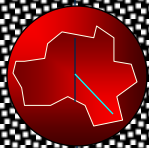
## **QC is different**

Entanglement and superposition provide a fascinating, powerful and fast way to compute.

## **QC is hard**

Designing quantum algorithms is very difficult.

Designing quantum hardware is very, very difficult.



# Good QM References

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*Book:*

**Modern Quantum Mechanics.**

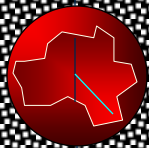
J.J. Sakurai,

Revised Edition. Addison-Wesley, 1994.

*This book is also translated into Korean:*

**양자역학.**

안동완 (역)



# Good QC References

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*Book:*

**Quantum Computing and Quantum Information.**

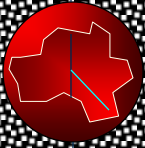
M. Nielsen and I. Chuang,  
Cambridge University Press, 2000.

*Online Lecture Notes:*

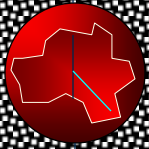
**Quantum Computation and Information.**

J. Preskill, (Cal Tech, Physics 229)

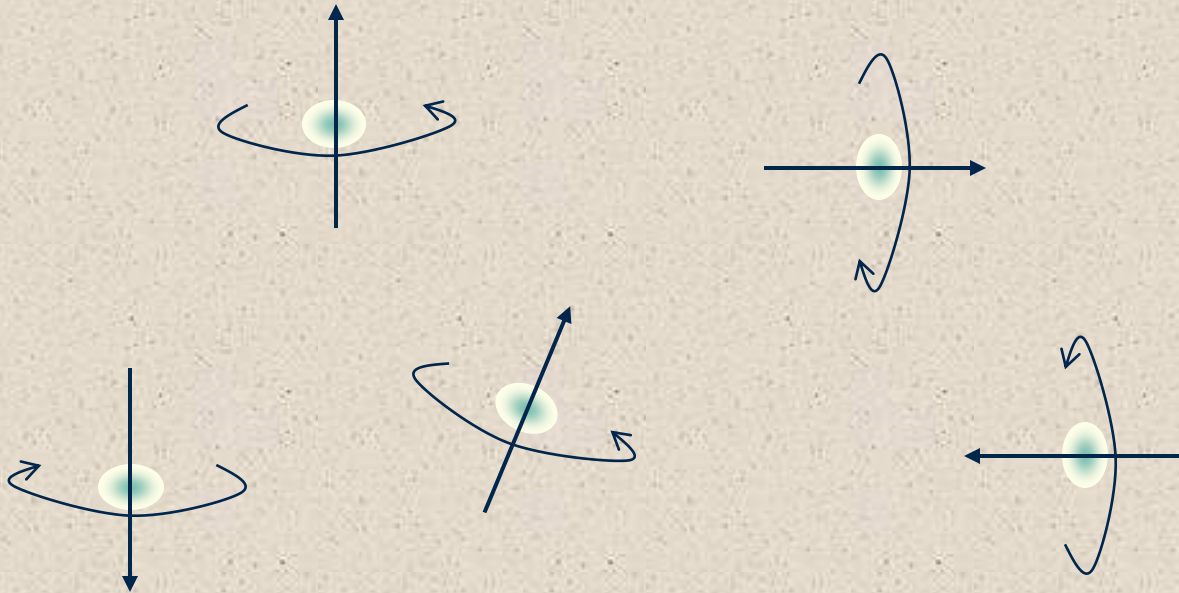
<http://www.theory.caltech.edu/people/preskill/ph229> [2000]



# Appendix Spinor Physics

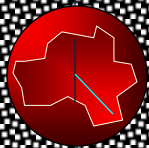


# Where do spinors come from?

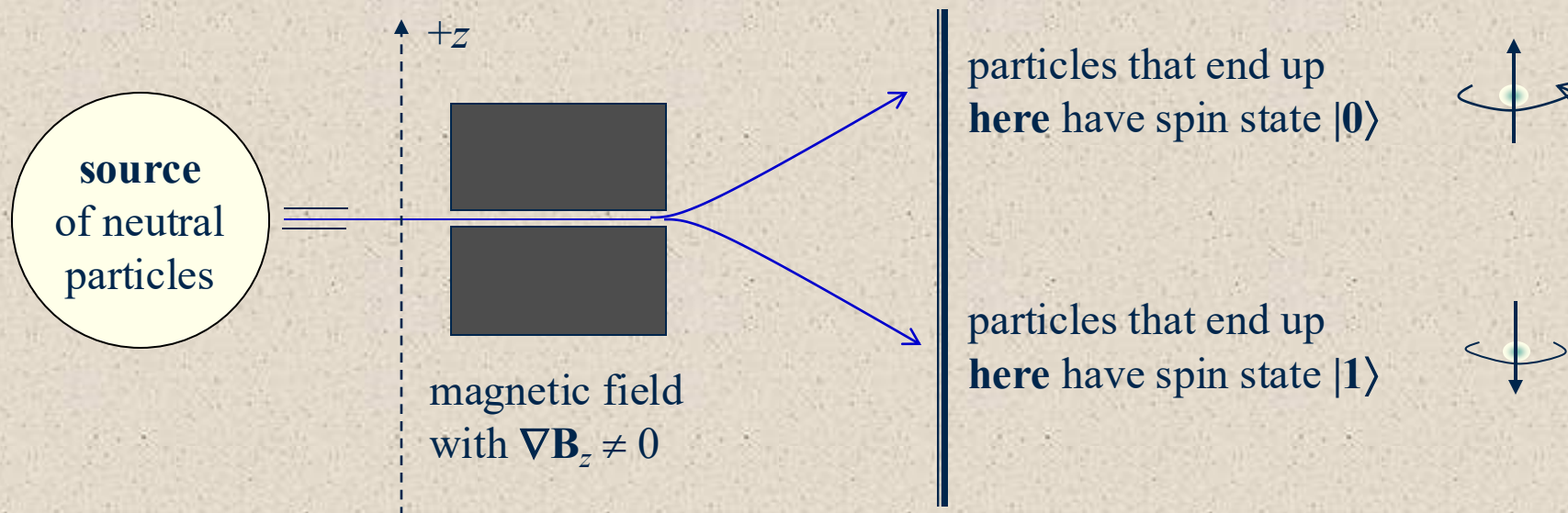


electrons, protons, neutrons  
atoms

Spinors model particles  
with **two** observed levels  
of “internal” spin

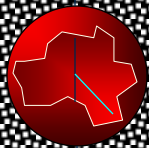


# Observing spin-1/2 particles



$|0\rangle$  = “spin UP in  $z$ -direction”

$|1\rangle$  = “spin DOWN in  $z$ -direction”

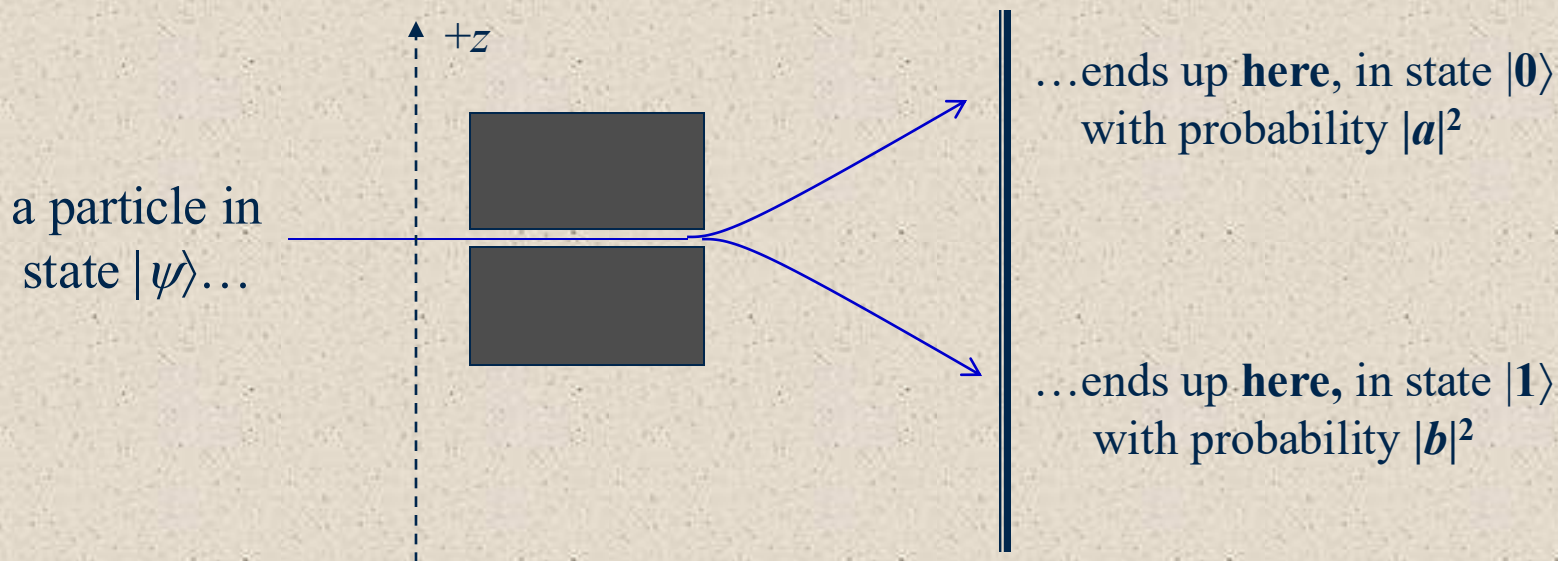


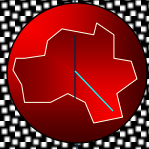
# States can be superposed

A particle can be in a **superposition** of these two states:

$$|\psi\rangle = a |0\rangle + b |1\rangle$$

with  $|a|^2 + |b|^2 = 1$





# Measurement

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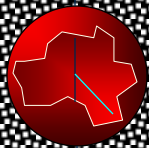
In quantum mechanical language, suppose someone is in the state:

$$0.5 \left| \text{☺} \right\rangle + 0.5 \left| \text{☹} \right\rangle$$

If you measure this person's **Mood**, then it they will be

- totally **happy** (☺) with a 50% chance
- totally **sad** (☹) with a 50% chance.

You will not find anything else (like ☹, “half-happy, half-sad”).



# Need complex numbers too...

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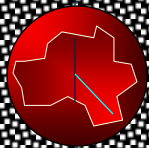
In quantum mechanical language, suppose someone is in the state:

$$[(1+4i)/\sqrt{30}] |\text{☺}\rangle + [(-3+2i)/\sqrt{30}] |\text{☹}\rangle$$

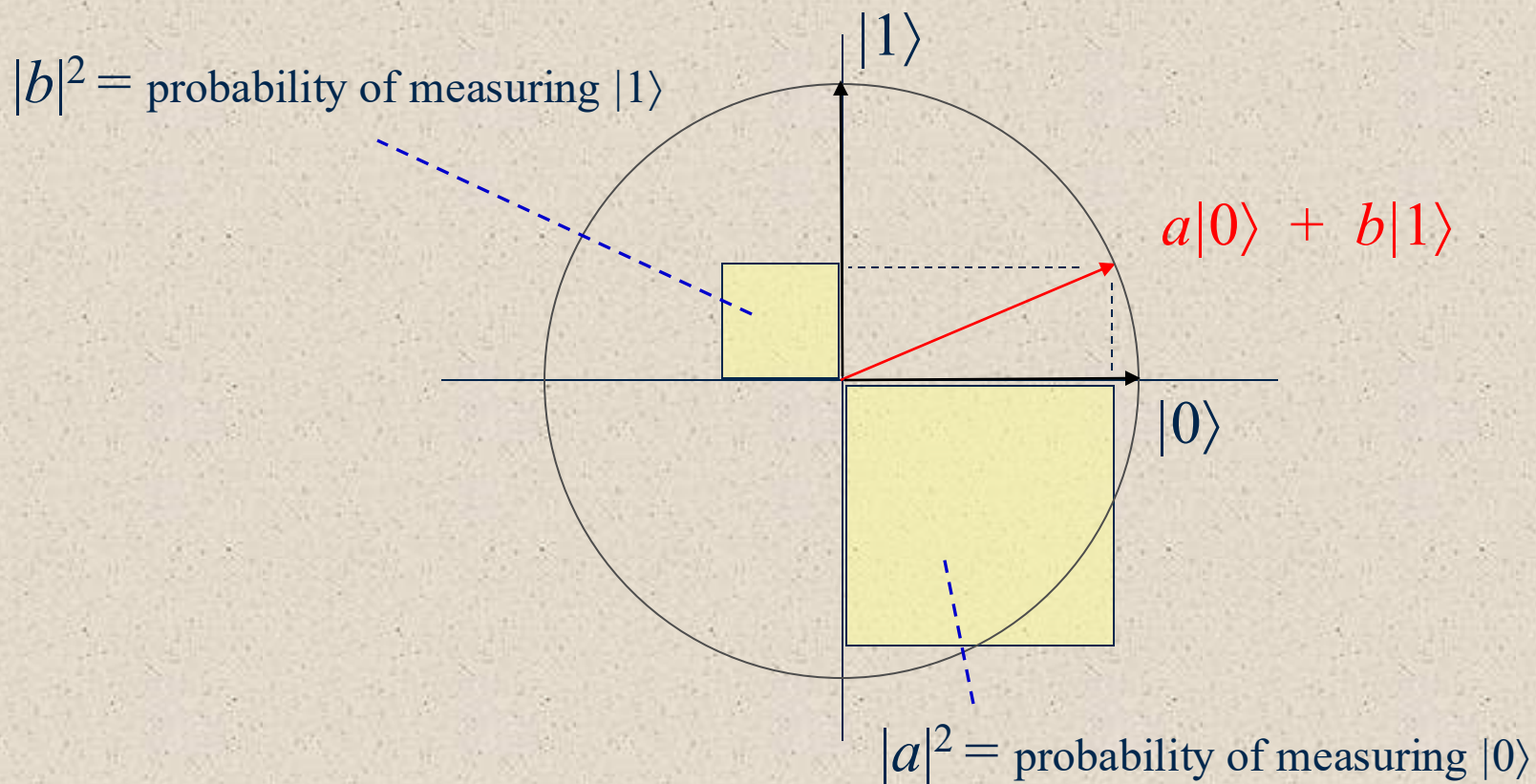
If you measure this person's **Mood**, then it they will be

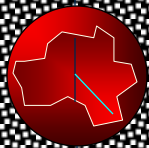
- happy (☺) with a  $| (1+4i)/\sqrt{30} |^2 = 57\%$  chance
- sad (☹) with a  $| (-3+2i)/\sqrt{30} |^2 = 43\%$  chance

You will not find anything else (like ☹, "half-happy, half-sad").



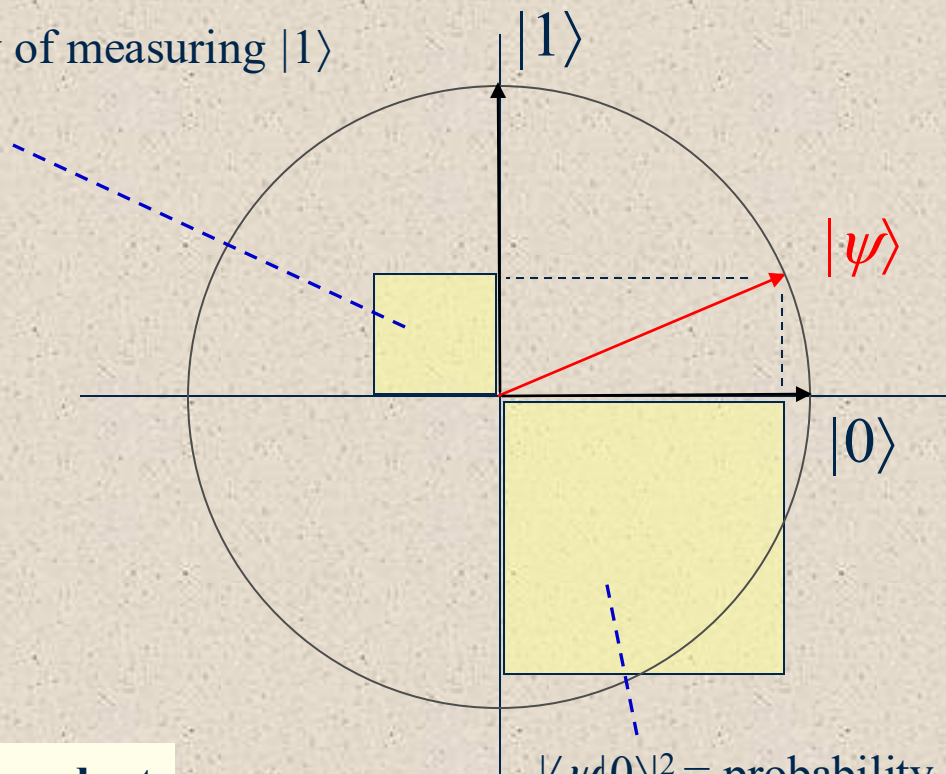
# Probabilities from states - 1





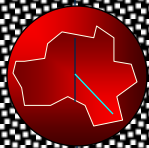
# Probabilities from states - 2

$|\langle\psi|1\rangle|^2 = \text{probability of measuring } |1\rangle$



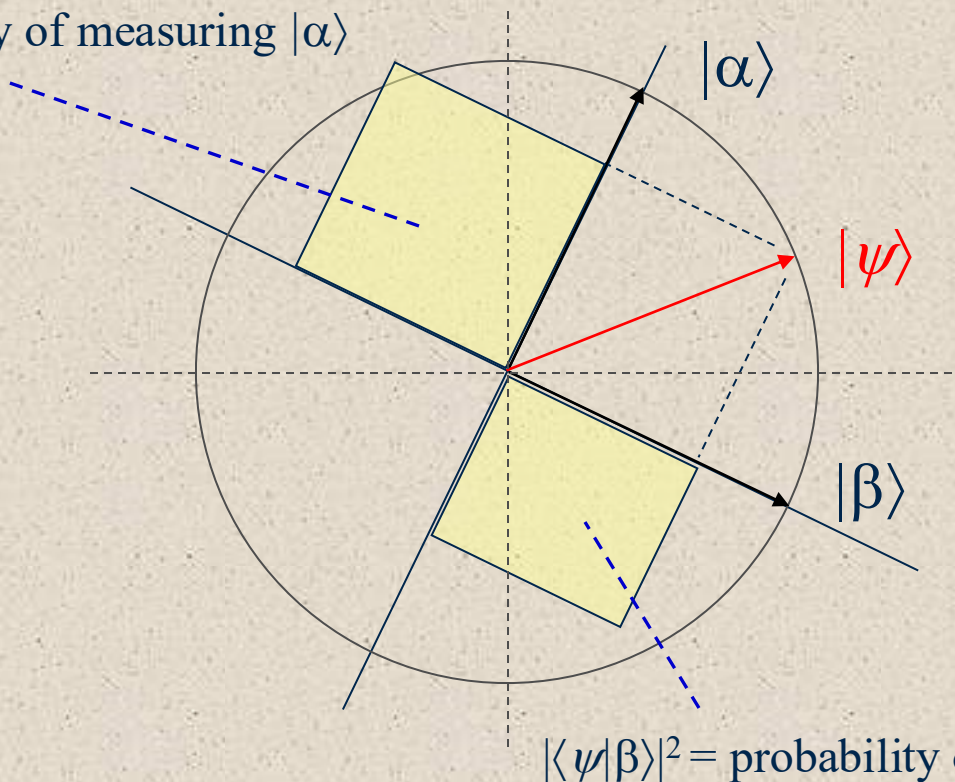
$\langle x|y\rangle$  is the **inner product**

$|\langle\psi|0\rangle|^2 = \text{probability of measuring } |0\rangle$



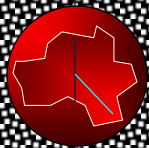
# Choose your measurement basis

$|\langle\psi|\alpha\rangle|^2 = \text{probability of measuring } |\alpha\rangle$

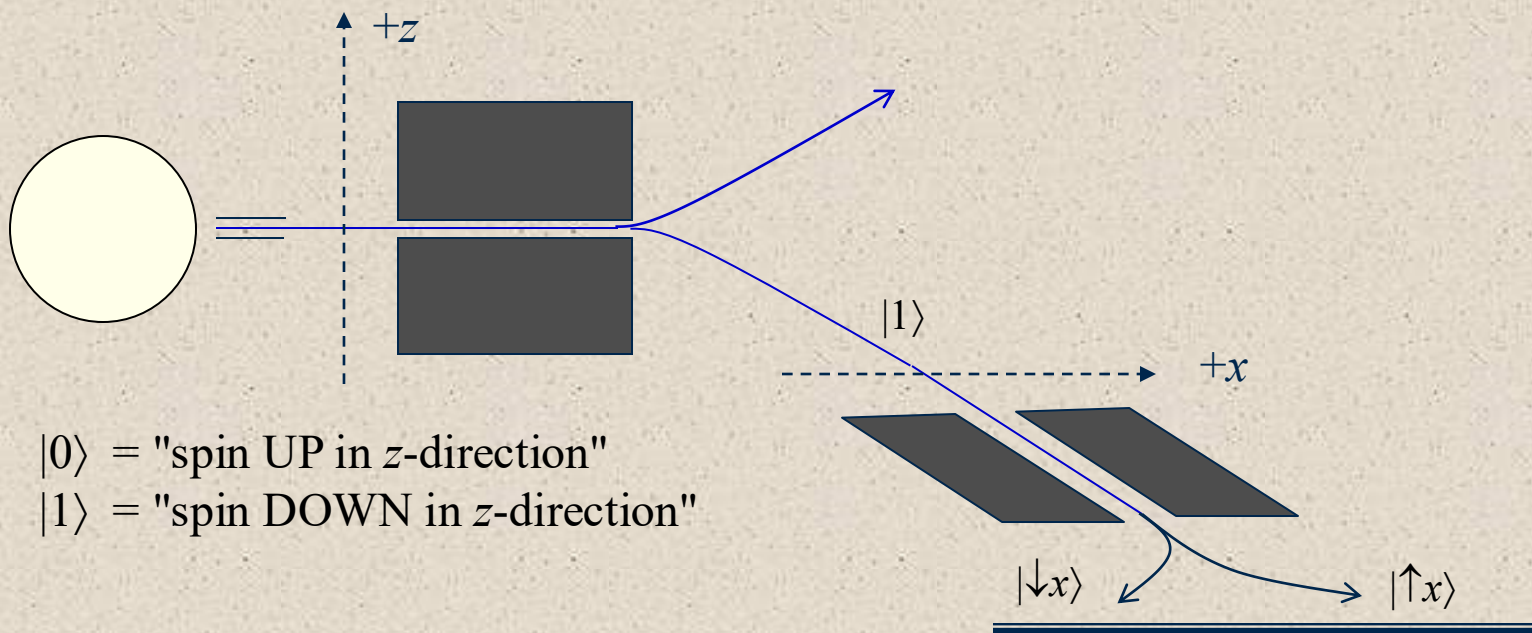


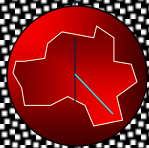
same state,  
different  
measurements

$|\langle\psi|\beta\rangle|^2 = \text{probability of measuring } |\beta\rangle$

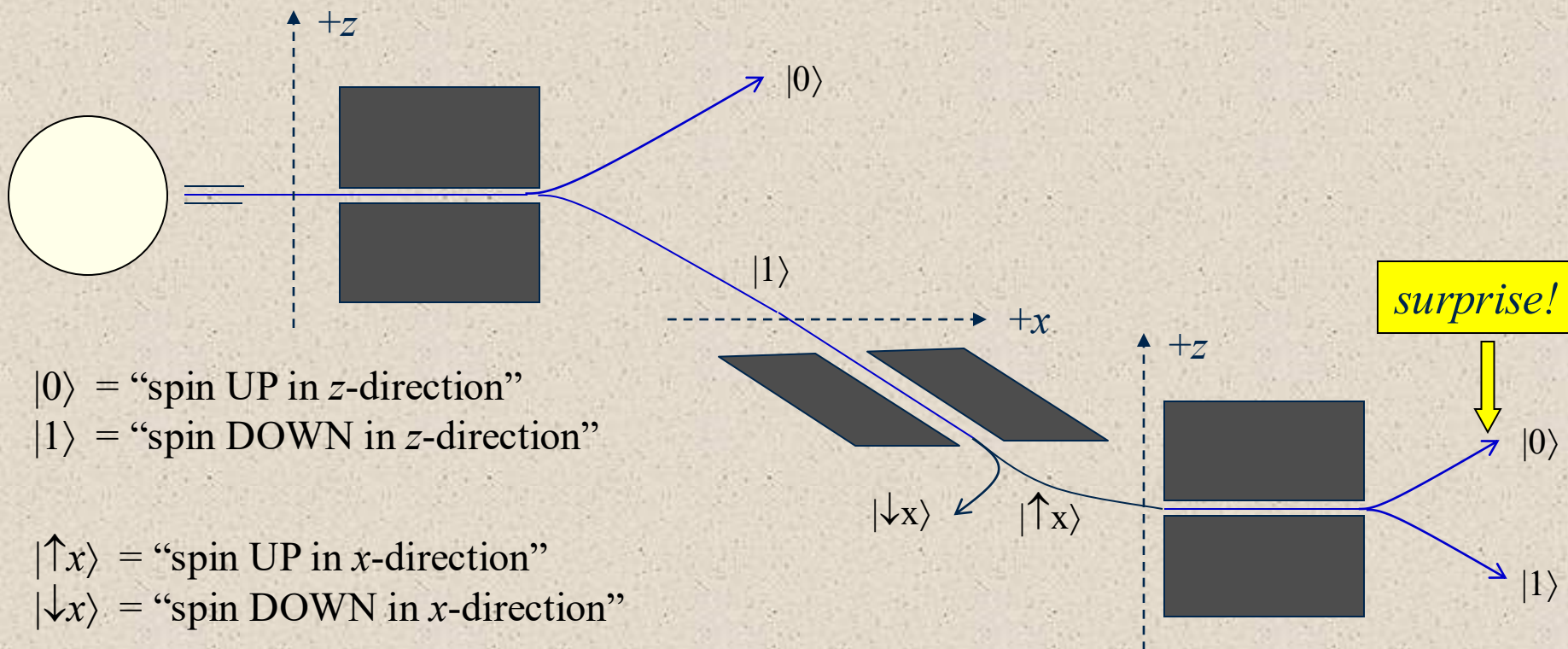


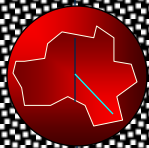
# Different bases - 1



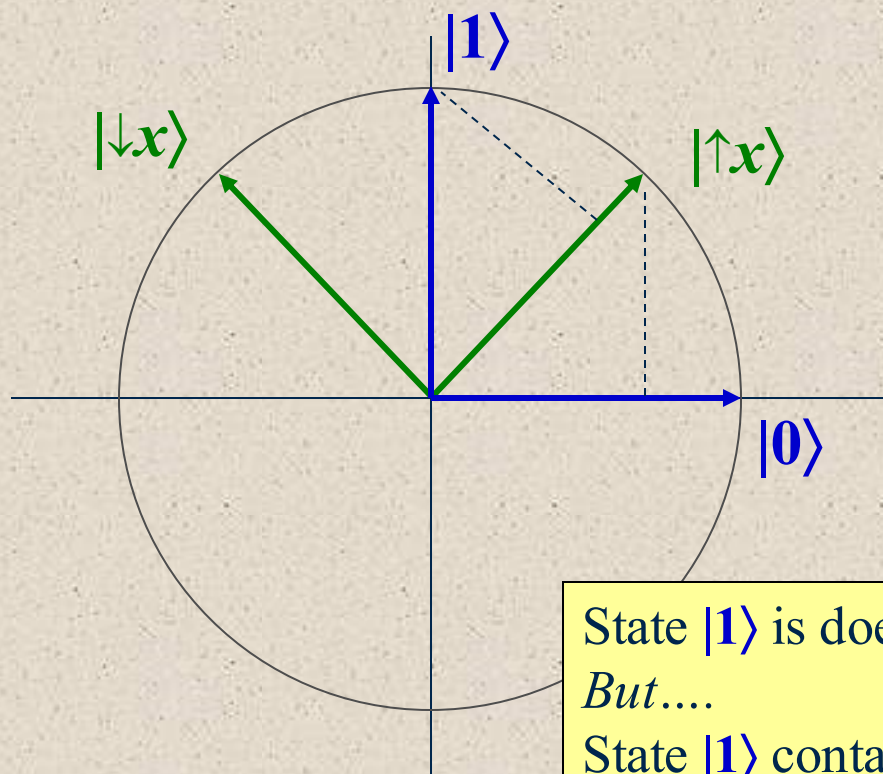


# Different bases - 2





# How it works....



$$|\uparrow x\rangle = 1/\sqrt{2} |0\rangle + 1/\sqrt{2} |1\rangle$$

$$|\downarrow x\rangle = 1/\sqrt{2} |0\rangle - 1/\sqrt{2} |1\rangle$$

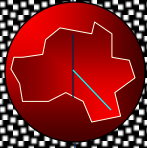
State  $|1\rangle$  does not contain any part of state  $|0\rangle$ .

*But....*

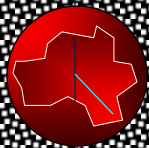
State  $|1\rangle$  contains a little bit of state  $|\uparrow x\rangle$ .

State  $|\uparrow x\rangle$  contains a little bit of state  $|0\rangle$ .

So, when talking about superpositions, “contains” is not transitive.

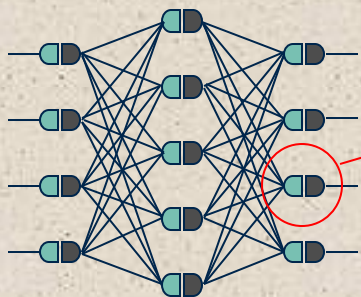


# Appendix Compared to Other “Paradigms”

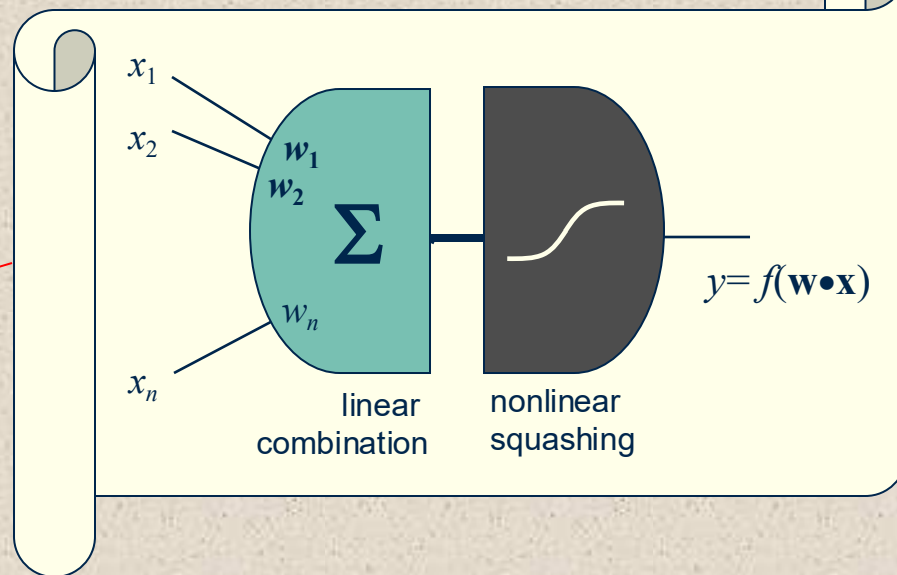


## Some Other “New Paradigms” - 1

# Neural Computing



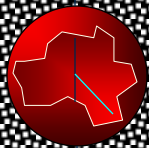
EXAMPLE



A collection of matrix-based algorithms that

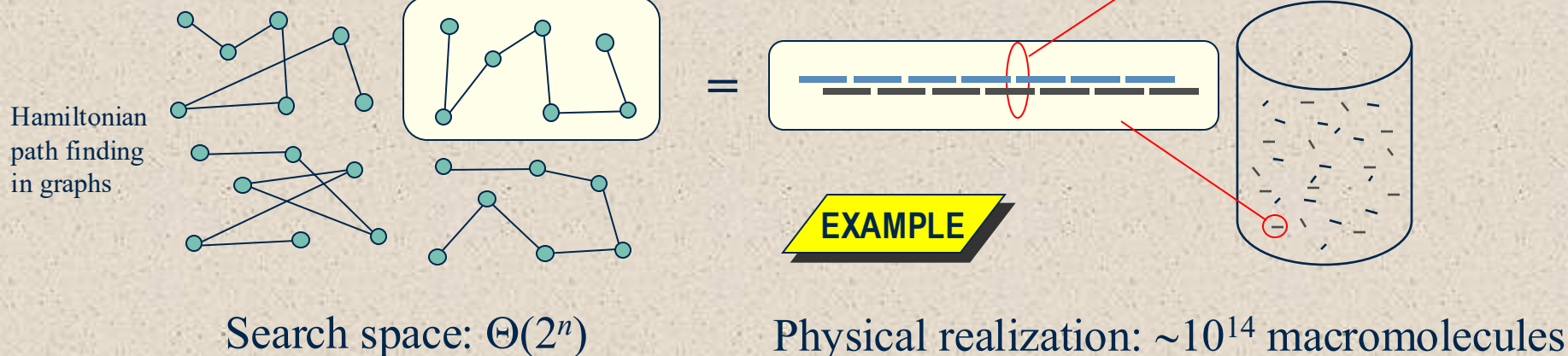
- allow parallel hardware implementation using highly interconnected simple processors;
- are sometimes useful in pattern recognition and control applications;
- are roughly based on a biological metaphor (*computing in the nervous system*)

**Not a fundamental change in the foundations of computer science.**



## Some Other “New Paradigms” - 2

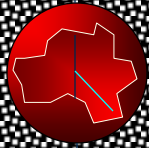
### Biomolecular Computing



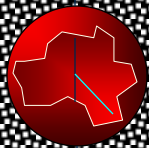
A massively parallel *physical realization* of pattern matching.

Large numbers of macromolecules interact using thermal energy.

**Not a fundamental change in the foundations of computer science.**



# Extra Slides



# Bases in tensor products

$\mathbb{F}^2$  is used in QC, not  $\mathbb{F}^3$ , but since  $2+2=2 \times 2$ , the  $\mathbb{F}^3$  case is better for clarifying tensor products.

You get a basis for the tensor product space  $\mathbb{F}^3 \otimes \mathbb{F}^3$  by forming **all** products of the original basis vectors:

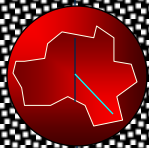
$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

**Define** Which, of course, is just the usual basis for  $\mathbb{F}^9$   
 $\mathbf{e}_{00} = \mathbf{e}_0 \otimes \mathbf{e}_0 =$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

**Define**  
 $\mathbf{e}_{ij} = \mathbf{e}_i \otimes \mathbf{e}_j.$

**This basis is**  
 $(\mathbf{e}_{00}, \mathbf{e}_{01}, \mathbf{e}_{02}, \dots, \mathbf{e}_{22})$



# Schmidt decomposition

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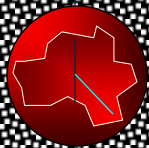
Using linear algebra tricks, we can re-express a general 2-qubit string

$$h_{00} \mathbf{e}_{00} + h_{01} \mathbf{e}_{01} + h_{10} \mathbf{e}_{10} + h_{11} \mathbf{e}_{11}$$

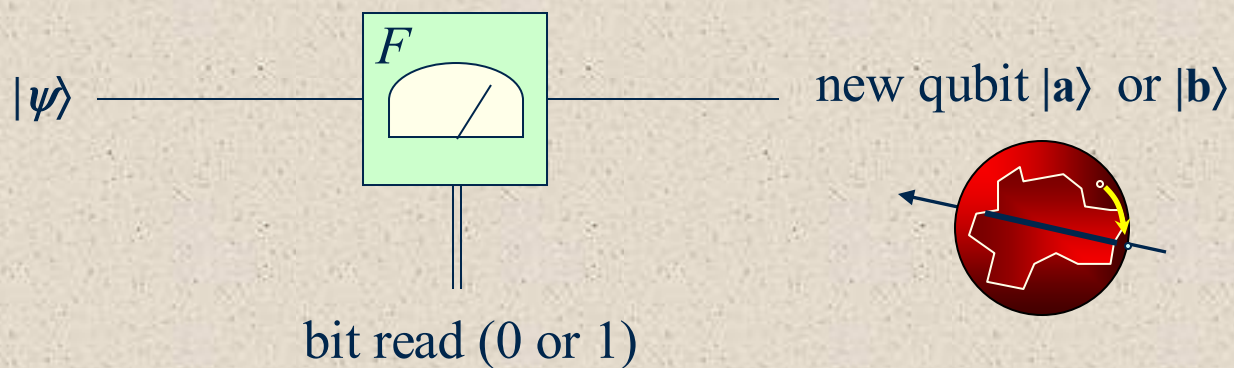
by changing the basis of each qubit so that we get the simple form:

$$s_0 \mathbf{f}_0 \otimes \mathbf{g}_0 + s_1 \mathbf{f}_1 \otimes \mathbf{g}_1.$$

(With 3 or more qubits, entanglement cannot be re-expressed so simply.)

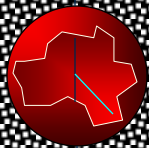


# Gate notation



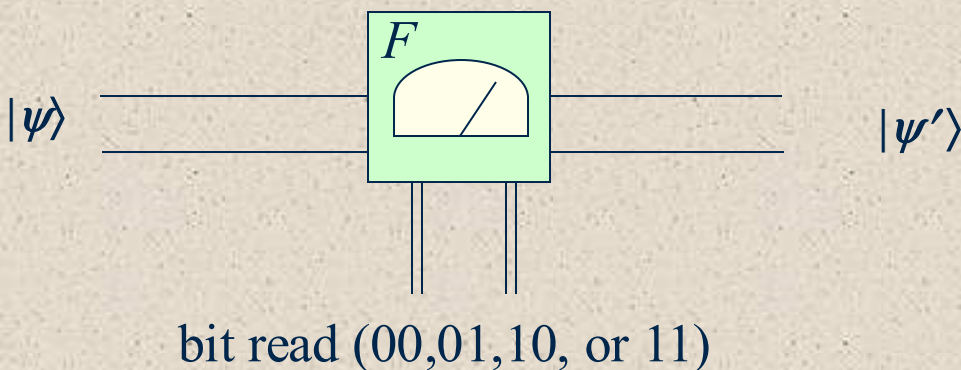
$F$  is an orthonormal basis (  $|a\rangle$  ,  $|b\rangle$  ) in  $\mathbb{R}^2$ .

If label  $F$  is omitted, basis is (  $|0\rangle$  ,  $|1\rangle$  )



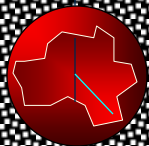
# Gate notation

*Measuring 2-qubits simultaneously.*

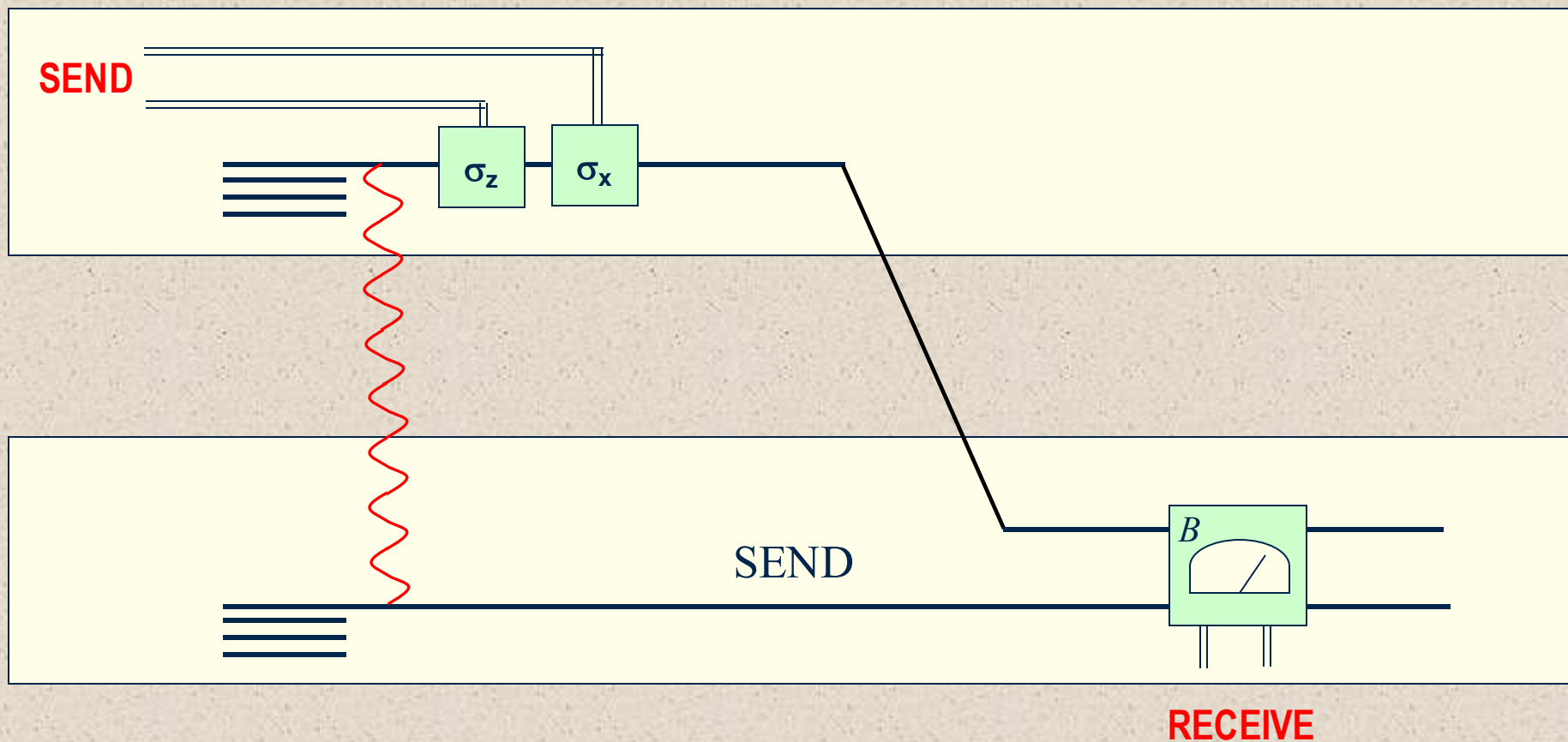


$F$  is an orthonormal basis (  $|a\rangle, |b\rangle$  ) in  $\mathbb{R}^2 \otimes \mathbb{R}^2$ .

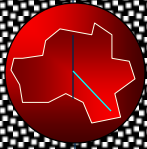
If label  $F$  is omitted, basis is (  $|00\rangle, |01\rangle, |10\rangle, |11\rangle$  ) .



## Superdense coding: *Sending 2 bits with 1 qubit*



$B$  is “Bell basis”: (  $|00\rangle + |11\rangle$ ,  $|00\rangle - |11\rangle$ ,  $|10\rangle + |01\rangle$ ,  $|10\rangle - |01\rangle$  )



# The Problem <sup>(1)</sup>

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## Problem

Given...

an set of records  $R = \{r_1, r_2, \dots r_N\}$ ;

a function  $f: R \rightarrow \{0,1\}$  such that  $f(r) = 0$  for all but 1 record;

Suppose...

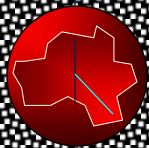
$R$  is represented as an unstructured list.

The function  $f$  takes  $O(1)$  time to evaluate on any input.

How long will it take to find  $f^{-1}(1)$  ?

## Solution

$O(N)$



# Complexity Conjectures

